



Framework Reference

Information Framework (SID)

Concepts and Principles

Information Framework Suite
GB922 Concepts and Principles
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Executive Summary

The Framework (formerly known as NGOSS) knowledge base contains a number of elements, including

- Business processes
- Business process flows
- Business application contracts
- Shared data and information
- Reference code and interfaces.

This guidebook focuses on the shared data and information elements of the Framework knowledge base. These two elements are commonly referred to as the Information Framework (also known as the SID) in this guidebook.

Framework knowledge elements can be viewed in many different ways as shown in the figure below. A business process analyst may be interested in a business view of the knowledge elements. For example, the definition of a business entity used by a process. From a systems point of view, a contract designer may desire to know the operations that act on a business entity. From an implementation point of view a component builder may be interested in a view of the realized information model.

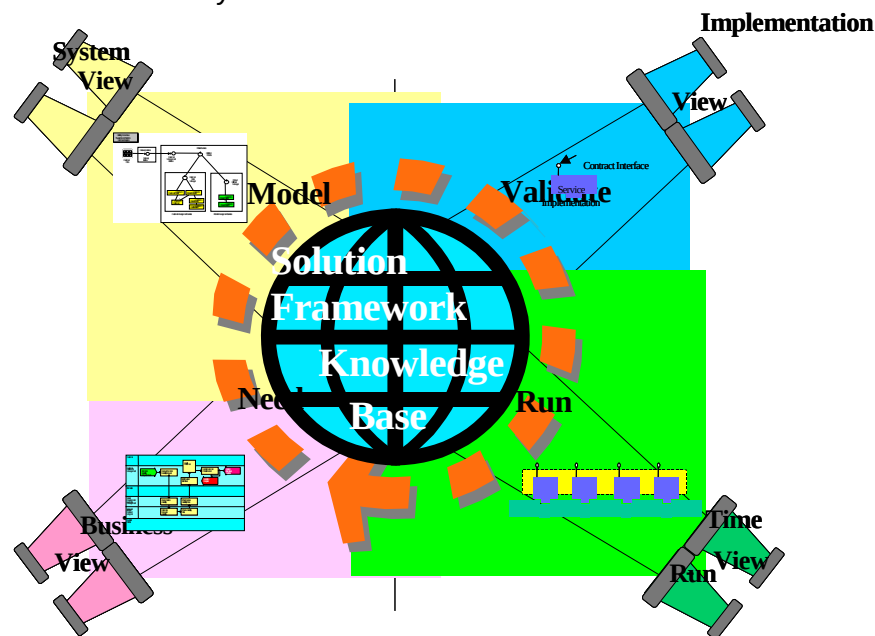


Figure 1 – Views of the Framework Knowledge Base

This document, Information Framework, Concepts and Principles, covers the following topics:

- Concepts and definitions key to understanding Information Framework business view content

- Information Framework concepts used to model and organize Information Framework business view content.

The detailed Information Framework business view content can be found in the associated GB922 Addenda.

This guidebook provides an insight into the organizing framework for the contents of the Information Framework business view, one of the three views referred to above.

A similar guidebook for the implementation view has been deemed to be out of scope for this version of the Information Framework deliverable.

The Information Framework not only satisfies the Framework information and data needs, but also satisfies another need – a mechanism to bring the business and IT communities into a closer understanding.

For many years the Business Process Framework (formerly known as the eTOM), and its predecessor the Telecom Operations Map (TOM), have provided a business process reference framework and common business process vocabulary. This framework and vocabulary have provided the communications and information industry enterprises an effective way to organize their business processes and communicate with each other.

The Information Framework (aka. SID) business view model can be viewed as a companion model to the Business Process Framework, in that it provides an information/data reference model and a common information/data vocabulary from a business entity perspective. The business view model uses the concepts of domains and aggregate business entities (or sub-domains) to categorize business entities, so as to reduce duplication and overlap. Based on data affinity concepts, the categorization scheme is necessarily layered, with each layer identifying in more detail the “things” associated with the immediate parent layer. This partitioning of the Information Framework business view model also allows distributed work teams to build out the model definitions while minimizing the flow-on impacts across the model.

Teamed with the Business Process Framework, the Information Framework provides enterprises with not only a process view of their business but also an entity view. That is to say, the Information Framework provides the definition of the ‘things’ that are to be affected by the business processes defined in the Business Process Framework. The Information Framework and Business Process Framework in combination offer a way to explain ‘how’ things are intended to fit together to meet a given business need. It should be noted that while both the Business Process Framework and the Information Framework business view model are layered, there is not necessarily a one-one relationship between the layers in each model, i.e., Business Process Framework Level 3 process elements do not only have relationships assigned to Information Framework Level 3 ABEs.

The relationship between the Information Framework and the Business Process Framework is outlined in more detail in Chapter 4. However, in brief, two sorts of relationships are identified – primary and secondary.

The primary relationship identifies the element of the Business Process Framework, which is responsible for creating instances of the Information Framework business entity. The presumption is that only one primary relationship exists between any Information Framework business entity and a Business Process Framework process element. In other words an enterprise should only use one process to create and

delete instances of a specific business entity, to reduce the risks of misaligned and/or non-unique information within the enterprise. This relationship underpins the concept of the single database of record or master database.

Secondary relationships are assigned between Business Process Framework process elements that use the information contained in the instantiated Information Framework business entities as part of their operation. There can be many secondary relationships identified, and this simply represents the reality that many enterprise processes rely on the same shared information, i.e. customer information is used by many business processes in different ways, but without changing the source information.

1. Introduction

1.1.Information Framework Business View Overview

The Information Framework business view addresses the information and communication service industry's need for shared information/data definitions and models. The definitions in the business view focus on business entity definitions and associated attribute definitions. A business entity is a thing of interest to the business, while its attributes are facts that further describe the entity. Together the definitions provide a business-oriented perspective of the information and data. When combined with business oriented UML class models, the definitions provide the business view of the information and data.

The content in the Information Framework Business View is organized using the Information Framework Model. The Information Framework was developed by the application of data affinity concepts to an enterprise's processes and data to derive a non-redundant view of the enterprise's, shared information and data. The result of this analysis is a layered framework, which partitions the shared information and data.

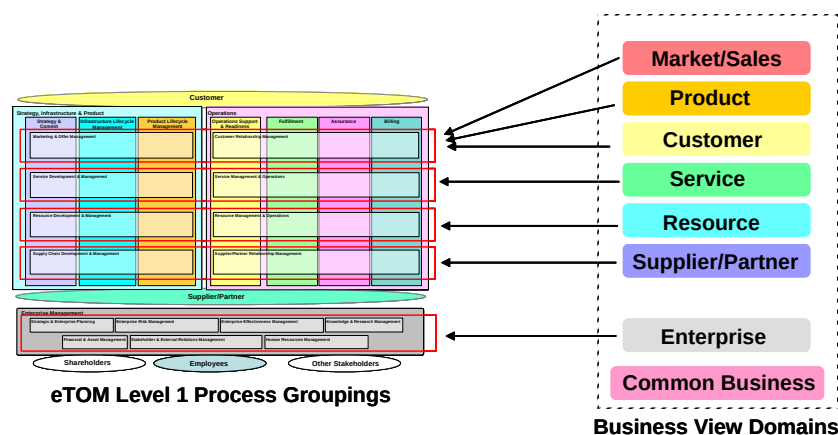


Figure 2 – Information Framework Domains & Business Process Framework Process Linkages

At the top layer, a set of domains is identified which are broadly aligned with the Business Process Framework business process framework as shown in Figure 2. Within each domain there is a high degree of cohesion between the identified business entities, and loose coupling between different domains. This enables segmentation of the total business problem and allows resources to be focused on a particular domain of interest. It is envisioned that the use of the resultant business entity definitions within each domain, when used in conjunction with the Business Process Framework, will provide a business view of the shared information and data.

Within each domain, further partitioning of the information is achieved through the identification of Aggregate Business Entities (ABE's). Figure 3 shows the currently identified Level 1 ABE's. As the Information Framework business view is further

expanded and defined, further partitioning of the ABE's occurs as more explicit business entities are identified.

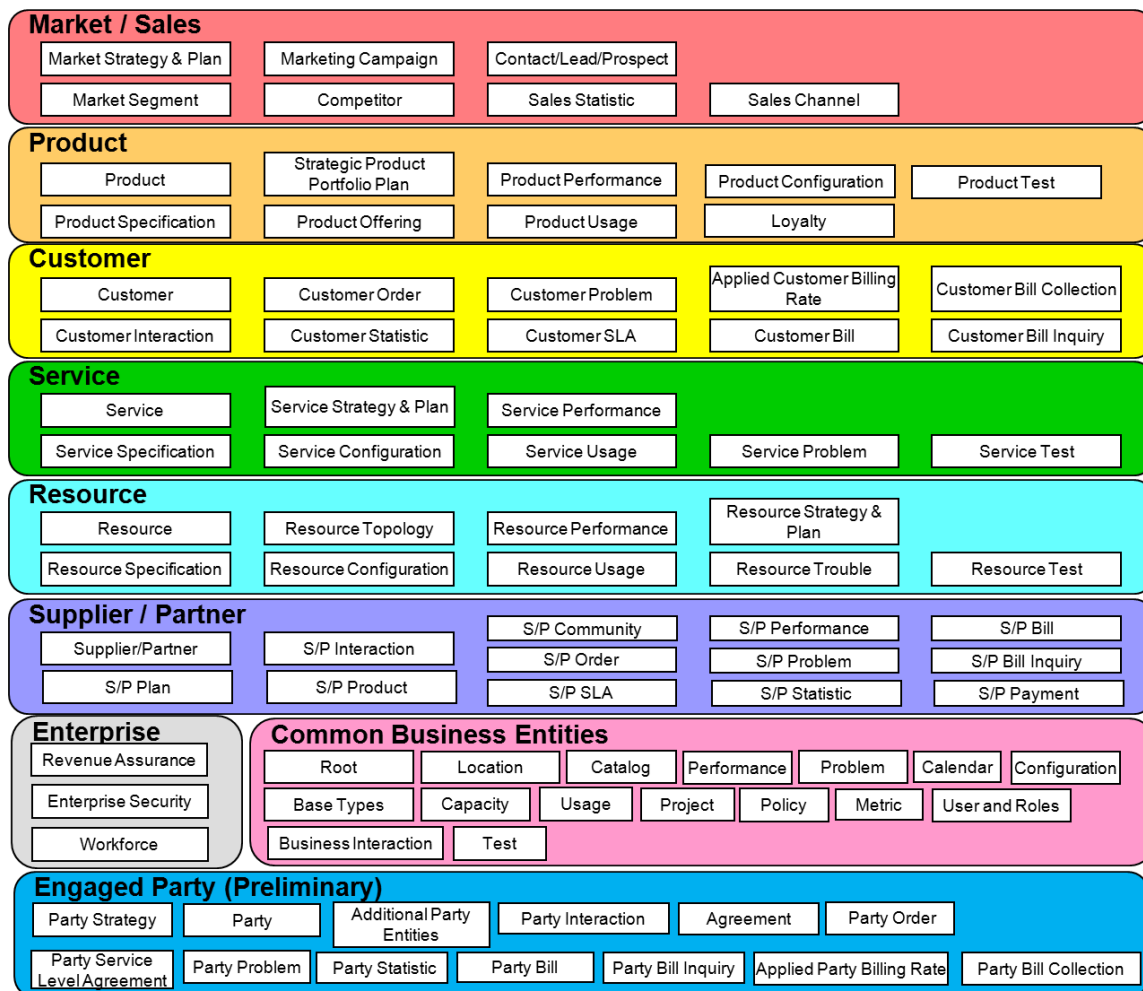


Figure 3 – Information Framework Domains & Level1 ABEs

The business entities along with the attributes and relationships that characterize the entities provide a view of the model that is easily understood from a business perspective. The business entities, attributes, and relationships are presented progressively developed using textual descriptions in each Information Framework addendum and in a consolidated Rational Rose UML-based model. The UML model provides an architecturally oriented business view of business entities, their attributes, and relationships to other business entities.

All domains are not yet covered in the Information Framework. As at the current revision of this document, only a subset of the identified domains has been defined. Domains included in this release are: Customer, Product (Product and Product Instance), Service, Resource (Physical and Logical), and Common Business Entities, such as Party, Location, Business Interaction, and Agreement that cross more than one domain. Domains not covered in this release include: Sales, Marketing, Supplier/Partner and Enterprise. Addenda to this document contain definitions of the business entities within each of these domains.

Domains and business entities contained in the Information Framework will expand as the Information Framework project progresses. Within each domain, TM Forum project needs were used to help scope the business entities defined and modeled in this document. The other domains and further definition of the five included domains will be presented in subsequent versions of this document or in other Information Framework documents.

The sources for the Information Framework include a variety of industry models, as well as models contributed by TM Forum member organizations. Where time permitted the contents of the Information Framework was mapped to the source models. Complete synthesis of the content of all models to find a common term for a concept was not possible. A best attempt was made to list cross-references to source models and synonyms for terms as part of the definition of the Information Framework business entities.

1.2. Definitions

Domain

A **Domain** is a collection of Aggregate Business Entities associated with a specific management area. Domains that make up the Information Framework are consistent with Business Process Framework level 0 concepts.

Domains are derived from an analysis of Process and Information Frameworks and have the following properties:

- Contain Business Entities that encapsulate both operations and corporate/enterprise information
- Are relatively stable collections of corporate/enterprise data and associated operations (in comparison with processes)
- Provide for robustness of corporate/enterprise data formats
- Provide clear responsibility and ownership.

Aggregate Business Entity

An **Aggregate Business Entity (ABE)** is a well-defined set of information and operations that characterize a highly cohesive, loosely coupled set of business entities.

Business Entity

A **Business Entity** represents something of interest to the business that may be tangible things (such as a Customer), active things (such as a Customer Order), or conceptual things (such as a Customer Account). Business entities are characterized by attributes and participate in relationships with other business entities. Business entity instances typically move through a well-defined life cycle.

Attribute

An **attribute** is a fact that describes a business entity.

Relationship

A **relationship** is an association of business interest between two business entities, or between a business entity and itself.

1.3. Information Framework Documents

The Information Framework business view model set of documents consists of a series of guide book that can be found on the framework's web page. A list of guide books can also be found in the framework's release notes.

1.4. Relationship to other TM Forum Documents

The relationships that the Information Framework document has with a number of other TM Forum documents are shown below:

Document	Scope	Purpose
Business Process Framework (GB921) (enhanced Telecom Operations Map)	Specifies service, network, element, and business management processes that are candidates for implementation using process flow components	Provides a high-level process framework describing typical transaction and data flows.

Figure 4- The relationship to other TM Forum documents

2. The Information Framework

2.1. Introduction

This chapter describes the Information Framework that is used to organize the Information Framework model. The Information Framework provides a high-level view into the Information Framework entities as well as providing an organizing structure in which the Information Framework business entities reside.

Figure 5 below shows how the domains contained within the Information Framework align with Business Process Framework level one domains/concepts. Whether taking a process or information perspective, it is important to be viewing the same set of concepts. The alignment is also a necessary enabler when mapping Business Process Framework processes to Information Framework business entities.

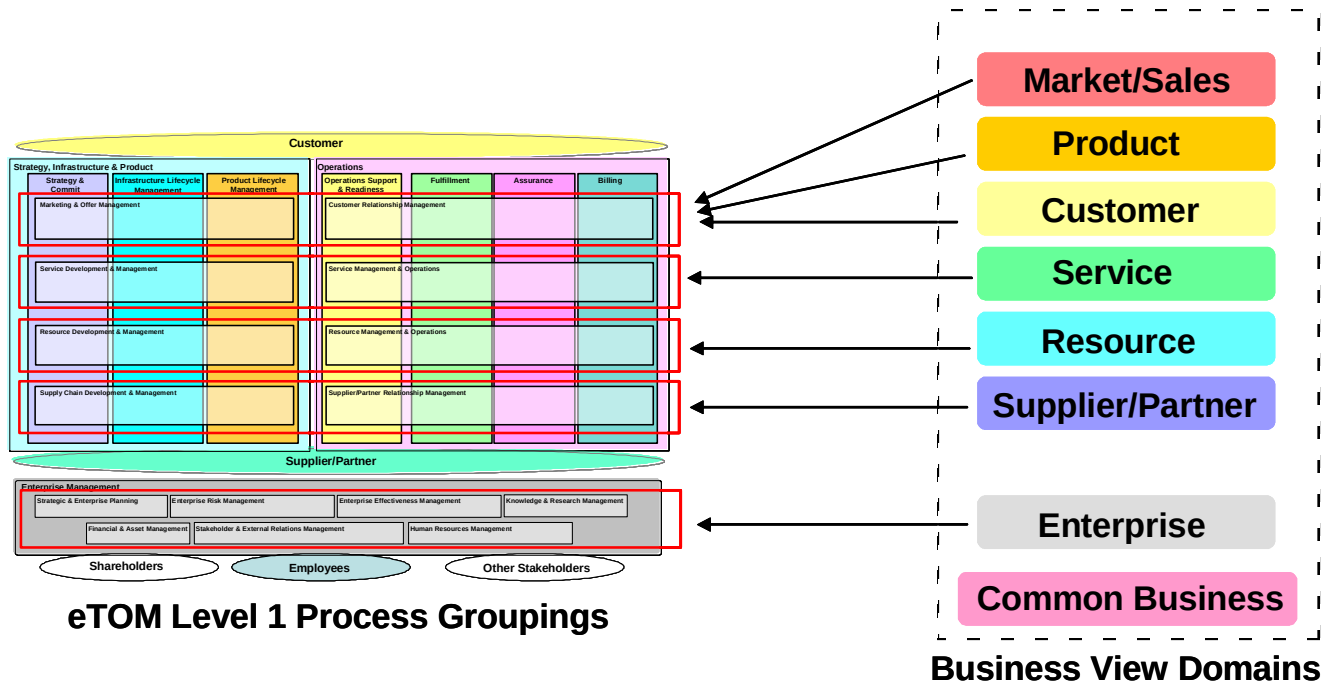


Figure 5- Business Process Framework/Information Framework Concepts/Domains

2.2. Information Framework – Level 1 ABEs

Figure 6 shows the Information Framework. The framework depicts the domains and Level 1 aggregate business entities (ABE) contained within each domain.

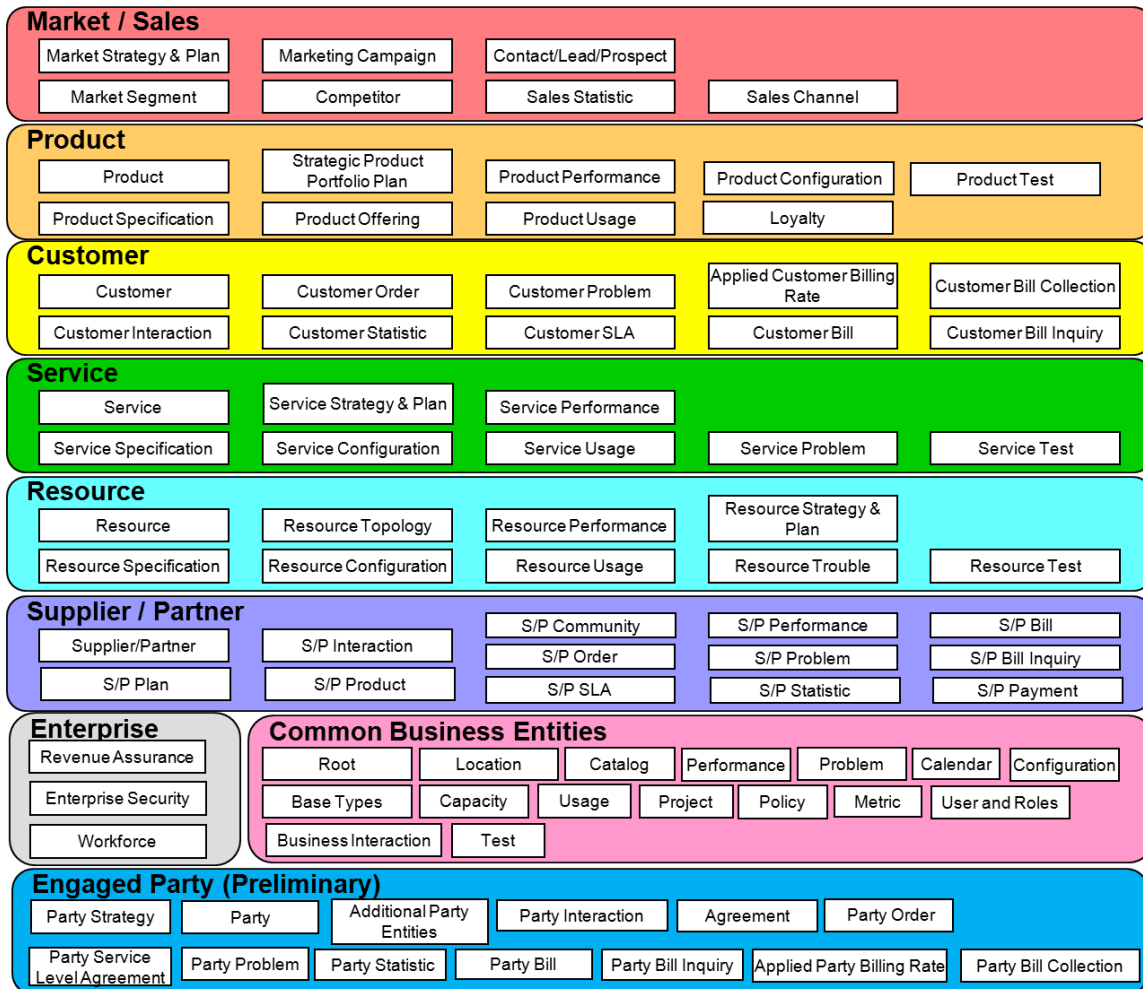


Figure 6 – Information Framework Business View Framework

As the development of the Information Framework has progressed, subsequent levels of ABEs have been identified. As an example, **Error! Reference source not found.** below shows Level 2 ABEs identified within the Market/Sales domain.

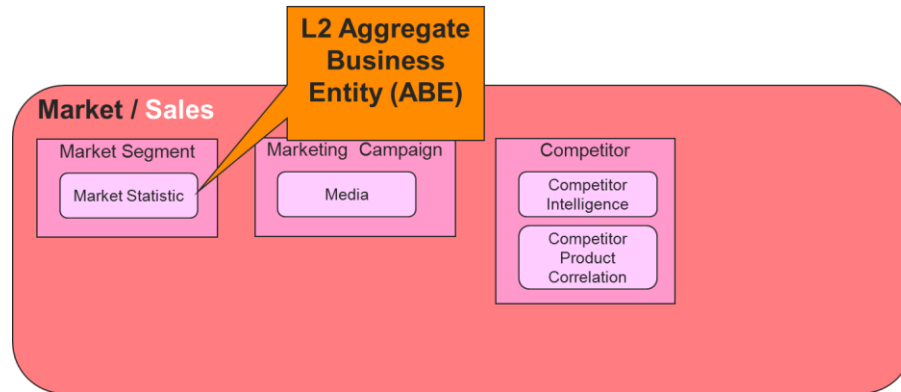


Figure 7 – Market/Sales Level 2 ABEs

The number of levels into which a Level 1 ABE decomposes is dependent on its complexity. Some Level 1 ABEs may decompose to two or more lower levels.

2.3. ABE Categorization

The ABE content and structure within each domain should be somewhat consistent. To ensure this, each ABE is aligned with a categorization pattern as described below. The pattern can also be used to confirm the completeness of each domain's ABE.

ABE categories include:

- Strategy and Plan
- Managed Entity
- Managed Entity Specification – A description of a ManagedEntity that might allow it to be built.
- Interaction – A communication with a ManagedEntity. This is a type of BusinessInteraction.
- Configuration – The internal structure of a ManagedEntity.
- Performance – The measure of ManagedEntity quality.
- Test – A means of interrogating a ManagedEntity in order to understand its state(s).
- Trouble – A problem associated with a ManagedEntity. Alarms, Outages and Faults are examples
- Price – The cost of a ManagedEntity.
- Usage – A period of time during which a ManagedEntity is in use.

Note: ABEs in Market/Sales and Common Business Domains have not yet been categorized.

	ABE Category									
	Managed Entity ABE			Management Entity ABE						
	Strategy and Plan	Managed Entity	Managed Entity Specification	Interaction	Configuration	Performance	Test	Trouble	Financial	Usage
Customer		Customer	Customer SLA	Customer Interaction, Customer Bill Inquiry	Customer Order	Customer Statistic,		Customer Problem	Customer Bill Collection, Applied Customer Billing Rate	Customer Bill
Product	Strategic Product Portfolio Plan	Product	Product Specification		Product Offering	Product Performance			Product Price (Product Offering Level 2)	Product Usage Statistic
Service	Service Strategy and Plan	Service	Service Specification		Service Configuration	Service Performance	Service Test	Service Problem		Service Usage
Resource	Resource Strategy and Plan	Resource	Resource Specification		Resource Configuration, Resource Topology	Resource Performance	Resource Test	Resource Trouble		Resource Usage
Supplier/ Partner	Supplier / Partner Plan	Supplier / Partner, Supplier Partner Product	S/P SLA	Supplier / Partner Interaction, S/P Bill Inquiry	S/P Order	Supplier / Partner Performance, S/P Statistic		S/P Problem	S/P Payment	

Figure 8 – Information Frame

work Common Categories

Notes: 1. The absence of an ABE in a particular category means either that the domain does not exactly fit the pattern or that an ABE has not yet been identified for the category.

3. Relationship to Business Process Framework

3.1. Domain definitions, ABE Definitions, and Mapping to Business Process Framework Level 2 Processes

3.1.1. Scope and Context

This section provides definitions for domains and ABEs, as well as a mapping of Information Framework ABEs to Business Process Framework Level 2 processes. It is based on the following sources:

- Information Framework/Business Process Framework Cross-referencing – Coupling of Information Framework-components to Business Process Framework level 2 processes, November 27, 2002, Roland Asztalos, Thomas Tenevall, Ki Consulting.
- Information Framework Business Process Framework Mapping (incomplete).xls, Wayne Sigley, Telstra.
- ABE Definitions for the Policy, Service, and Resource Domains, June 2003, John Strassner.
- Comments and input from core Information Framework team, notably John Reilly and John Strassner.

3.1.2. Principles for Mapping

The following principles and delimitations are used for coupling Information Framework ABEs to Business Process Framework business processes.

- When possible, an ABE is coupled to only one level 2 Business Process Framework process. This specific process is denoted as “primary”.
- Primary level 2 Business Process Framework processes manage the complete life cycle of an ABE, by creating, reading, updating, and deleting (CRUD) entity instances contained within the ABE.
- Information is considered to flow both in the process layer as well as the systems (read ABE) layer
- If a one-to-one pairing is not attainable, an ABE is mapped to at most two level 2 Business Process Framework processes. If an ABE is mapped to two level 2 Business Process Framework processes, both level 2 processes are considered as primary.
- Processes requiring information feeds from ABEs or read an ABE's entities to function properly are termed as “secondary”.

3.1.3. Regarding ABEs Supporting Two Level 2 Business Process Framework Processes

Some ABEs are coupled to two or more level 2 Business Process Framework processes. This occurs because of as the result of:

- *Duality phenomenon*: implies that an arbitrary ABE supports two primary level 2 Business Process Framework processes, whereas the two primary processes belong to separate process areas. Thus, three combinations are available, [1] Operations & SIP, [2] Operations & EM or [3] SIP & EM. The duality phenomenon takes place because the ABE in question is regarded as unclear in terms of “core functionality”, since it does not explicitly support only one primary level 2 Business Process Framework process.
- *Ambiguity phenomenon*: is similar to the duality phenomenon, however, here an arbitrary ABE supports two primary level 2 Business Process Framework processes which both belong to only one of the three available process areas.

These phenomena may have any number of causes, such as

- The ABE may be separated into two ABEs, each supporting a single level 2 Business Process Framework process.
- There may be a missing ABE that supports one of the level 2 Business Process Framework processes.
- There may be a missing Business Process Framework level 2 process that aggregates the two processes.
- A mistake could have been made in the mapping.

The phenomena will be investigated and, if possible, resolved in subsequent phases of the Information Framework project via the continued collaboration between the Information Framework team and the Business Process Framework team.

3.1.4. Market/Sales Domain

The Market/Sales Domain includes data and contract operations that support the sales and marketing activities needed to gain business from customers and potential customers. On the Sales side, this includes sales contacts/leads/prospects through to the sales-force and sales statistics. Market includes market strategy and plans, market segments, competitors and their products, through to campaign formulation.			
Market/Sales ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	ExampleSecondary Business Process Framework Core Processes
Market Segment Supports market segments, market statistics, and forecasts.	M&OM – SC	Market Strategy & Policy	Sales Development
			Product & Offer Portfolio Planning
			Product & Offer Development & Retirement
			Supply Chain Strategy & Planning

The Market/Sales Domain includes data and contract operations that support the sales and marketing activities needed to gain business from customers and potential customers. On the Sales side, this includes sales contacts/leads/prospects through to the sales-force and sales statistics. Market includes market strategy and plans, market segments, competitors and their products, through to campaign formulation.			
Market/Sales ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	ExampleSecondary Business Process Framework Core Processes
Market Strategy and Plan Supports the business plans and strategies on how to address the market with appropriate products and channels.	M&OM – SC	Market Strategy & Policy	Resource Strategy & Planning
			Supply Chain Strategy & Planning
			Product & Offer Portfolio Planning
Competitor Identifies other providers who compete in the same market segments, accumulates intelligence about the competitors, including products (price, Key Performance Indicators and so forth).	M&OM – SC	Market Strategy & Policy	Product & Offer Portfolio Planning
			Enterprise Performance Assessment
Marketing Campaign Supports marketing new or existing product offerings to identified target markets. For example, the launch of a pre-paid product with multiple promotions across distribution channels, market segments and so forth; a new campaign for an existing product; a re-launch of a campaign for an existing product.	M&OM – PLM	Product Marketing Communications & Promotion	CRM Support & Readiness
			Marketing Fulfillment Response
			Marketing Capability Delivery
Sales Statistics Maintains sales forecasts, new service requirements, customer needs, and customer education, as well as calculating key performance indicators about Sales & Marketing revenue and sales channel performance.	CRM – OSR	CRM Support & Readiness	Market Strategy & Policy
			Product & Offer Portfolio Planning
			Sales Development
Sales Channel Keeps track of distribution channels and sales activities, sales quotas, sales contests, commission/bonus plans, commissions/bonuses, and maintains groups of individuals that make up the sales force.	M&OM – PLM	Sales Development	S/P Performance Management
			S/P Settlements & Billing Management
			Marketing Capability Delivery
			CRM Support & Readiness

The **Market/Sales** Domain includes data and contract operations that support the sales and marketing activities needed to gain business from customers and potential customers. On the Sales side, this includes sales contacts/leads/prospects through to the sales-force and sales statistics. Market includes market strategy and plans, market segments, competitors and their products, through to campaign formulation.

Market/Sales ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	ExampleSecondary Business Process Framework Core Processes
Contact/Lead/Prospect Provides the ability to track sales leads through their life cycle up until the time the leads become customers, including lead and contact information, sales prospects, proposals made to potential customers, and the amount of potential revenue the leads represent in the form of a sales pipeline.	CRM – F	Marketing Fulfillment Response^	
		Selling^	
Notes: ^ Due to ABE ambiguity			

3.1.5. Product Domain

The **Product** Domain is concerned with the lifecycle of products and information and contract operations related to products' lifecycle. The Domain contains Aggregate Business Entities that deal with the strategic portfolio plans, products offered, product performance, product usage statistics, as well as the product instances delivered to a customer.

Product ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	Example Secondary Business Process Framework Core Processes
Strategic Product Portfolio Plan Is concerned with the plans of the product portfolio, which product offerings to make available to each market segment and the plans to development and deploy product offerings, as well as retirement of products.	M&OM – SC	Product & Offer Portfolio Planning	Market Strategy & Policy
			Product & Offer Development & Retirement
Product Specification Defines the functionality and characteristics of product offerings made available to the market.	M&OM – PLM	Product & Offer Development & Retirement	Product & Offer Capability Delivery
			Service Configuration & Activation
			SM&O Support & Readiness

The Product Domain is concerned with the lifecycle of products and information and contract operations related to products' lifecycle. The Domain contains Aggregate Business Entities that deal with the strategic portfolio plans, products offered, product performance, product usage statistics, as well as the product instances delivered to a customer.			
Product ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	Example Secondary Business Process Framework Core Processes
Product Offering Represents tangible and intangible goods and services made available for a certain price to the market in the form of product catalogs. This ABE is also responsible for targeting market segments based on the appropriate market strategy.	M&OM – PLM	Product & Offer Development & Retirement	Marketing Fulfillment Response
			Product Marketing Communications & Promotion
			Product & Offer Capability Delivery
			Selling
			Order Handling
Product Represents an instance of a product offering subscribed to by a party, such as a customer, the place where the product is in use, as well as configuration characteristics, such as assigned telephone numbers and internet addresses. The Product ABE also tracks the services and/or resources through which the product is realized.	CRM – F	Order Handling	Selling
			Service Configuration & Activation
			Problem Handling
			Customer QoS/SLA Management
			Billing & Collections Management
			Service Problem Management
			Service Quality Management
			Service Guiding & Mediation
Product Configuration The definition of how a Product operates or functions in terms of CharacteristicSpecification(s) and related ResourceSpec(s), ProductSpec(s), ServiceSpec(s) as well as a representation of how a Product operates or functions in terms of characteristics and related Resource(s), Product(s), Service(s).			SM&O Support & Readiness

The Product Domain is concerned with the lifecycle of products and information and contract operations related to products' lifecycle. The Domain contains Aggregate Business Entities that deal with the strategic portfolio plans, products offered, product performance, product usage statistics, as well as the product instances delivered to a customer.			
Product ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	Example Secondary Business Process Framework Core Processes
Product Usage Represents usage of products associated with Customers used for charging that may appear on a Customer Bill.	CRM – OSR	CRM Support & Readiness	Product & Offer Development & Retirement
			Problem Handling
			Customer QoS/SLA Management
			Enterprise Performance Assessment
Product Performance The Product Performance ABE handles product performance goals, the results of end-to-end product performance assessments, and the comparison of assessments against goals. The results may include the identification of potential capacity issues.	CRM – OSR	CRM Support & Readiness	Product & Offer Development & Retirement
			Customer QoS/SLA Management
			Service Quality Management
			Resource Performance Management
			Enterprise Performance Assessment
Loyalty A loyalty program is one of the tools used by the loyalty process to retain customers. The Loyalty ABE contains all entities useful to specify and instantiate loyalty programs. A LoyaltyProgramProdSpec defines the LoyaltyRules that have to be checked in order to identify the actions to apply. Depending on the type of LoyaltyRules a LoyaltyAccount might be needed to collect gains or not. A LoyaltyProgramProduct is a type of ProductComponent and described by a LoyaltyProgramProdSpec.			
Product Test A product test is a function performed on a product that results in measures being produced that reflect the functioning of the entity under test.			

3.1.6. Customer Domain

The Customer Domain includes all data and contract operations associated with individuals or organizations that obtain products from an enterprise, such as a service provider. It represents of all types of contact with the customer, the management of the relationship, and the administration of customer data. The Customer Domain also includes data and contract operations related to the customer bills for products, collection of payment, overdue accounts, and the billing inquiries and adjustments made as a result of inquiries.			
Customer ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	Example Secondary Business Process Framework Core Processes
Customer Is the focus for the Customer domain. Customer data is the enterprise's knowledge of the customer and accounts held by a customer.	CRM – F	Selling	Product Marketing Communications & Promotion
			Order Handling
			Problem Handling
			Customer QoS/SLA Management
			Billing & Collections Management
			Retention & Loyalty
			Customer Interaction Management
Customer Interaction Represents communications with customers, and the translation of customer requests and inquiries into appropriate “events” such as the creation of an customer order, the creation of a customer bill inquiry, or the creation of a customer problem.	CRM – FAB	Customer Interaction Management	Selling
			Order Handling
			Problem Handling
			Customer QoS/SLA Management
			Billing & Collections Management
			Retention & Loyalty
Customer Statistic Represents the analysis of customer usage patterns, customer profitability statistics and churn and retention statistics.	CRM – FAB	Customer Management	CRM Support & Readiness
			Selling
			Order Handling
			Problem Handling
			Customer QoS/SLA Management

The Customer Domain includes all data and contract operations associated with individuals or organizations that obtain products from an enterprise, such as a service provider. It represents of all types of contact with the customer, the management of the relationship, and the administration of customer data. The Customer Domain also includes data and contract operations related to the customer bills for products, collection of payment, overdue accounts, and the billing inquiries and adjustments made as a result of inquiries.			
Customer ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	Example Secondary Business Process Framework Core Processes
			Billing & Collections Management
Customer Problem Focuses on technical assistance and problem handling for customers.	CRM – A	Problem Handling	Retention & Loyalty
Customer SLA Is a special case of the Service Level Agreement ABE where an involved party in the agreement is a Customer. See the Agreement ABE in the Engaged Party Domain for details.	CRM – A	Customer QoS/SLA Management	Selling
			Retention & Loyalty
			Billing & Collections Management
Customer Order Handles single customer orders and the various types thereof, such as regulated and non-regulated orders.	CRM – F	Order Handling	Retention & Loyalty
			Service Configuration & Activation
Customer Bill Handles real-time and non-real-time Call Detail Records (CDRs) and other sources of data that result in invoice items. The Customer Bill ABE also represents the format of a bill, schedule the production of bills, customer invoicing profiles, all the financial calculations necessary to determine the total of the bill (except for rating and rating discounts), and credits and adjustments to bills.	CRM – B	Billing & Collections Management	Retention & Loyalty
			Service Guiding & Mediation
Applied Customer Billing Rates Deals with the correlation of related usage for subsequent rating, rates applied to the usage (both regulated and non-regulated), discounts to usage, and any taxes due on the rated usage.	CRM – B	Charging	Retention & Loyalty
			Billing & Collections Management
Customer Bill Collections Handles credit violations, actions for overdue debts, and facility billing audits.	CRM – B	Billing & Collections Management	Retention & Loyalty
			Financial Management

The **Customer** Domain includes all data and contract operations associated with individuals or organizations that obtain products from an enterprise, such as a service provider. It represents of all types of contact with the customer, the management of the relationship, and the administration of customer data. The Customer Domain also includes data and contract operations related to the customer bills for products, collection of payment, overdue accounts, and the billing inquiries and adjustments made as a result of inquiries.

Customer ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	Example Secondary Business Process Framework Core Processes
Customer Bill Inquiry Represents invoice inquiries associated with invoices sent to customers and handles disputes and adjustments on individual charges, invoices, and accounts.	CRM – B	Bill Inquiry Handling	Customer Management

3.1.7. Service Domain

The **Service** Domain consists of a set of layered ABEs that are used to manage the definition, development, and operational aspects of Services provided by a Framework system. Entities in this domain support various Business Process Framework processes that deal with the definition, development and management of services offered by an enterprise. This includes agreement on Service levels to be offered, deployment and configuration of Services, management of problems in Service installation, deployment, usage, or performance, quality analysis, and rating. Finally, this domain also includes entities to perform planning for future offerings, service enhancement or retirement, and capacity.

Service ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	Example Secondary Business Process Framework Core Processes
Service Specification The Service Specification ABE contains entities that define the invariant characteristics and behavior of both types of Service entities. This enables multiple instances to be derived from a single specification entity. In this derivation, each instance will use the invariant characteristics and behavior defined in its associated template. Entities in this ABE focus on adherence to standards, distinguishing features of a Service, dependencies (both physical and logical, as well as on other services), quality, and cost. In general, entities in this ABE enable Services to be bound to Products and run using Resources.	SD&M – PLM	Service Development & Retirement	Service Capability Delivery
			Service Configuration & Activation
			Service Problem Management
			Service Quality Management
			Service Guiding & Mediation
			SM&O Support & Readiness
Service The Service ABE contains entities that are used to	SM&O – F	Service Configuration & Activation	Service Capability Delivery

The **Service** Domain consists of a set of layered ABEs that are used to manage the definition, development, and operational aspects of Services provided by a Framework system. Entities in this domain support various Business Process Framework processes that deal with the definition, development and management of services offered by an enterprise. This includes agreement on Service levels to be offered, deployment and configuration of Services, management of problems in Service installation, deployment, usage, or performance, quality analysis, and rating. Finally, this domain also includes entities to perform planning for future offerings, service enhancement or retirement, and capacity.

Service ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	Example Secondary Business Process Framework Core Processes
represent both customer-facing and resource-facing types of services. Entities in this ABE provide different views to examine, analyze, configure, monitor and repair Services of all types. Entities in this ABE are derived from Service Specification entities.			Service Problem Management
			Service Development & Retirement
			SM&O Support & Readiness
			Service Quality Management
			Service Guiding & Mediation
			Service Capability Delivery
			Service Development & Retirement
			Service Quality Management
			Service Guiding & Mediation
Service Configuration The definition of how a Service operates or functions in terms of CharacteristicSpecification(s) and related ResourceSpec(s) and ServiceSpec(s)	SM&O – F	Service Configuration & Activation	Service Capability Delivery

The Service Domain consists of a set of layered ABEs that are used to manage the definition, development, and operational aspects of Services provided by a Framework system. Entities in this domain support various Business Process Framework processes that deal with the definition, development and management of services offered by an enterprise. This includes agreement on Service levels to be offered, deployment and configuration of Services, management of problems in Service installation, deployment, usage, or performance, quality analysis, and rating. Finally, this domain also includes entities to perform planning for future offerings, service enhancement or retirement, and capacity.			
Service ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	Example Secondary Business Process Framework Core Processes
as well as a representation of how a Service operates or functions in terms of characteristics and related Resource(s) and Service(s).			SM&O Support & Readiness
			Resource Provisioning
Service Performance The Service Performance ABE collects, correlates, consolidates, and validates various performance statistics and other operational characteristics of customer and resource facing service entities. It provides a set of entities that can monitor and report on performance. Each of these entities also conducts network performance assessment against planned goals, performs various aspects of trend analysis, including error rate and cause analysis and Service degradation. Entities in this ABE also manage the traffic generated by a Service, as well as traffic trend analysis. This is important for newer technologies that separate data, control and management functions for a given Service.	SM&O – A	Service Quality Management	Service Problem Management
			SM&O Support & Readiness
			Service Development & Retirement
			Resource Performance Management
Service Test A service test is a function performed on a service that results in measures being produced that reflect the functioning of the entity under test.	SM&O – F	Service Configuration & Activation	Service Problem Management
			Service Quality Management
Service Problem The Service Problem ABE manages faults, alarms, and outages from a Service point-of-view. This is then correlated to trouble tickets, regardless of whether the cause is physical or logical. Other entities in this ABE are used to direct the recovery from each of these three types of	SM&O – A	Service Problem Management	SM&O Support & Readiness
			Problem Handling

The **Service** Domain consists of a set of layered ABEs that are used to manage the definition, development, and operational aspects of Services provided by a Framework system. Entities in this domain support various Business Process Framework processes that deal with the definition, development and management of services offered by an enterprise. This includes agreement on Service levels to be offered, deployment and configuration of Services, management of problems in Service installation, deployment, usage, or performance, quality analysis, and rating. Finally, this domain also includes entities to perform planning for future offerings, service enhancement or retirement, and capacity.

Service ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	Example Secondary Business Process Framework Core Processes
problems. They provide the ability to associate Resource faults and alarms to degradation and outages of Services that run on those Resources. These functions are independent of the Resources and technologies used to build the Service. A third set of entities in this ABE is used to differentiate between customer-reported problems and network-induced problems.			Service Quality Management
			Resource Trouble Management
Service Usage The Service Usage ABE collects Service consumption data, and generates Service usage records, for use by other business entities. The entities in this ABE provide physical, logical, and network usage information.	SM&O – B	Service Guiding & Mediation	Resource Data Collection & Processing
			Service Quality Management
			Service Development & Retirement
Service Strategy & Plan The Service Strategy and Plan ABE contains entities that are used to address the need for enhanced or new Services, as well as the retirement of existing Services, by the enterprise. These entities have a strong dependency to both entities in the Resource and Product domains. Resulting efforts, such as deciding what Resources to use to host a Service, or what Services are used to support new Product Specifications, are also supported, as are service demand forecasts.	SD&M – SC	Service Strategy & Planning	Product & Offer Portfolio Planning
			Service Capability Delivery
			Service Development & Retirement
			Resource Strategy & Planning

3.1.8. Resource Domain

The **Resource** Domain consists of a set of layered ABEs that are used to manage the definition, development, and operational aspects of the information computing and processing infrastructure of a Framework system. It supports the Business Process Framework processes that deal with the definition, development and management of the infrastructure of an enterprise. This includes the components of the infrastructure as well as Products and Services that use this infrastructure.

The Resource Domain has three important objectives. The first is to associate Resources to Products and Services, and provide a detailed enough set of Resource entities (organized as ABEs) to facilitate this association. The second is to ensure that Resources can support and deliver Services offered by the enterprise. Management of resources involves planning, configuration, and monitoring to capture performance, usage, and security information. This also includes the ability to reconfigure Resources in order to fine tune performance, respond to faults, and correct operational deficiencies in the infrastructure. Resources also provide usage information which is subsequently aggregated to the customer level for billing purposes. The final objective of the Resource domain is to enable strategy and planning processes to be defined. Entities in the Resource domain may be associated with processes that involve planning new and/or enhanced Services, or even the retirement of Services, offered by the Enterprise.

Resource ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	Example Secondary Business Process Framework Core Processes
Resource Strategy & Plan The Resource Plan ABE is used to plan networks and resource elements both initially and for growth. It will coordinate both logical and physical resource growth. Inputs are budgets from business sources, service forecasts, current and projected network utilization, new technologies, and retiring technologies. It handles the lifecycle (installation, modification, removal, and retirement) for both logical and physical resources.	RD&M – SC	Resource Strategy & Planning	Resource Development & Retirement
			Service Strategy & Planning
			Resource Capability Delivery
			RM&O Support & Readiness
Resource Specification The Resource Specification ABE contains entities that define the invariant characteristics and behavior of each type of Resource entities. This enables multiple instances to be derived from a single specification entity. In this derivation, each instance will use the invariant characteristics and behavior defined in its associated template.	RD&M – PLM	Resource Development & Retirement	RM&O Support & Readiness
			Resource Provisioning
			Resource Capability Delivery
			Resource Trouble Management
			Resource Performance Management
			Resource Data Collection & Processing
Resource The Resource ABE contains entities that are used to represent the various aspects of a Resource. This includes four sets of entities that represent: the physical and logical aspects of a Resource; show	RM&O – F	Resource Provisioning	RM&O Support & Readiness
			Resource Trouble Management

The **Resource** Domain consists of a set of layered ABEs that are used to manage the definition, development, and operational aspects of the information computing and processing infrastructure of a Framework system. It supports the Business Process Framework processes that deal with the definition, development and management of the infrastructure of an enterprise. This includes the components of the infrastructure as well as Products and Services that use this infrastructure.

The Resource Domain has three important objectives. The first is to associate Resources to Products and Services, and provide a detailed enough set of Resource entities (organized as ABEs) to facilitate this association. The second is to ensure that Resources can support and deliver Services offered by the enterprise. Management of resources involves planning, configuration, and monitoring to capture performance, usage, and security information. This also includes the ability to reconfigure Resources in order to fine tune performance, respond to faults, and correct operational deficiencies in the infrastructure. Resources also provide usage information which is subsequently aggregated to the customer level for billing purposes. The final objective of the Resource domain is to enable strategy and planning processes to be defined. Entities in the Resource domain may be associated with processes that involve planning new and/or enhanced Services, or even the retirement of Services, offered by the Enterprise.

Resource ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	Example Secondary Business Process Framework Core Processes
how to aggregate such resources into aggregate entities that have physical and logical characteristics and behavior; and show how to represent networks, sub-networks, network components, and other related aspects of a network.			Resource Capability Delivery
			Resource Performance Management
			Resource Data Collection & Processing
Resource Topology The Resource Topology ABE contains entities that define physical, logical, and network topological information. This information is critical for assessing the current state of the network, as well as providing information on how to fix problems, tune performance, and in general work with the network (both as a whole and with its components). Each of these topological views provides its own physical, logical, or network related information that can be used to manage one or more layers in a layered network.	RD&M – ILM	Resource Capability Delivery	Resource Development & Retirement
			RM&O Support & Readiness
Resource Configuration The definition of how a Resource operates or functions in terms of CharacteristicSpecification(s) and related ResourceSpec(s), as well as a representation of how a Resource operates or functions in terms of characteristics and related Resource(s).	RM&O – F	Resource Provisioning	Resource Capability Delivery
			RM&O Support & Readiness
Resource Trouble The Resource Trouble ABE manages problems found in allocated resource instances, regardless of whether the problem is physical or	RM&O – A	Resource Trouble Management	RM&O Support & Readiness

The **Resource** Domain consists of a set of layered ABEs that are used to manage the definition, development, and operational aspects of the information computing and processing infrastructure of a Framework system. It supports the Business Process Framework processes that deal with the definition, development and management of the infrastructure of an enterprise. This includes the components of the infrastructure as well as Products and Services that use this infrastructure.

The Resource Domain has three important objectives. The first is to associate Resources to Products and Services, and provide a detailed enough set of Resource entities (organized as ABEs) to facilitate this association. The second is to ensure that Resources can support and deliver Services offered by the enterprise. Management of resources involves planning, configuration, and monitoring to capture performance, usage, and security information. This also includes the ability to reconfigure Resources in order to fine tune performance, respond to faults, and correct operational deficiencies in the infrastructure. Resources also provide usage information which is subsequently aggregated to the customer level for billing purposes. The final objective of the Resource domain is to enable strategy and planning processes to be defined. Entities in the Resource domain may be associated with processes that involve planning new and/or enhanced Services, or even the retirement of Services, offered by the Enterprise.

Resource ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	Example Secondary Business Process Framework Core Processes
logical. Entities in this ABE detect these problems, act to determine their root cause, resolve these problems and maintain a history of the activities involved in diagnosing and solving the problem. Detecting problems can be done via software (e.g. responding to an alarm) and/or by hardware (e.g. a measurement or probe) and/or manually (e.g. visual inspection). This includes tracking, reporting, assigning people to fix the problem, testing and verification, and overall administration of repair activities.			Resource Performance Management
			Service Problem Management
Resource Test A resource test is an execution of functionality on a resource that results in measures being produced.	RM&O – F	Resource Provisioning	Resource Capability Delivery
			Resource Trouble Management
			Resource Performance Management
Resource Performance The Resource Performance ABE collects, correlates, consolidates, and validates various performance statistics and other operational characteristics of Resource entities. It provides a set of entities that can monitor and report on performance. The entities in this ABE provide physical, logical, and performance information. Each of these entities also conducts network performance assessment against planned goals,	RM&O – AB	Resource Performance Management	RM&O Support & Readiness
			Resource Development & Retirement

The **Resource** Domain consists of a set of layered ABEs that are used to manage the definition, development, and operational aspects of the information computing and processing infrastructure of a Framework system. It supports the Business Process Framework processes that deal with the definition, development and management of the infrastructure of an enterprise. This includes the components of the infrastructure as well as Products and Services that use this infrastructure.

The Resource Domain has three important objectives. The first is to associate Resources to Products and Services, and provide a detailed enough set of Resource entities (organized as ABEs) to facilitate this association. The second is to ensure that Resources can support and deliver Services offered by the enterprise. Management of resources involves planning, configuration, and monitoring to capture performance, usage, and security information. This also includes the ability to reconfigure Resources in order to fine tune performance, respond to faults, and correct operational deficiencies in the infrastructure. Resources also provide usage information which is subsequently aggregated to the customer level for billing purposes. The final objective of the Resource domain is to enable strategy and planning processes to be defined. Entities in the Resource domain may be associated with processes that involve planning new and/or enhanced Services, or even the retirement of Services, offered by the Enterprise.

Resource ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	Example Secondary Business Process Framework Core Processes
performs various aspects of trend analysis, including error rate and cause analysis and Resource degradation. Entities in this ABE also manage traffic in a Resource. This includes statistics defining Resource loading, and traffic trend analysis.			Resource Data Collection & Processing
Resource Usage The Resource Usage ABE collects Resource consumption data, and generates Resource usage records, for use by other business entities. The entities in this ABE provide physical, logical, and network usage information.	RM&O – B	Resource Data Collection & Processing	Billing & Collections Management Service Guiding & Mediation

3.1.9. Supplier/Partner Domain

The SupplierPartner Domain includes all SupplierPartner-oriented data and contract operations associated with a SupplierPartner. Its scope encompasses, planning of strategies vs. SupplierPartners, handling of all types of contact with the SupplierPartner, the management of the relationship, and the administration of SupplierPartner data. The SupplierPartner Domain also includes data and contract operations related to the SupplierPartner Bills, disputes and inquiries.			
SupplierPartner ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	Example Secondary Business Process Framework Core Processes
SupplierPartner Plan The SupplierPartner Plan ABE handles the strategies and the planning of the business relation with the SupplierPartner with input from other ABEs, such as MarketSales, S/P Performance and Competitor Analysis.	SCD&M – SC	Supply Chain Strategy & Planning	Market Strategy & Policy
			Service Strategy & Planning
			Resource Strategy & Planning
			Sales Development
			Strategic Business Planning
			S/P Performance Management
			Business Development
			Enterprise Performance Assessment
SupplierPartner Is the focus for the SupplierPartner Domain. SupplierPartner represents the enterprise's knowledge of the SupplierPartner, their accounts and the relations the Enterprise has with the SupplierPartner. It also contains all Supplier Partner agreements and negotiations.	SCD&M – ILM and/or SCD&M – PLM and/or S/PRM –S&R	Supply Chain Capability Delivery [^] *	Business Development
		Supply Chain Development & Change Management [^] *	Service Configuration & Activation
		S/PRM Support & Readiness*	S/P Problem Reporting & Management
			S/P Settlements & Billing Management
SupplierPartner Interaction Represents communication with SupplierPartners, and translating SupplierPartner requests and inquiries into appropriate “events”. <i>Initiated by the Service Provider as a Customer</i> such as the requisition for the Wholesale Service, the quotation order for the End-Customer instance, the quotation update	S/PRM – FAB	S/P Interface Management	Supply Chain Development & Change Management
			S/PRM Support & Readiness

The SupplierPartner Domain includes all SupplierPartner-oriented data and contract operations associated with a SupplierPartner. Its scope encompasses, planning of strategies vs. SupplierPartners, handling of all types of contact with the SupplierPartner, the management of the relationship, and the administration of SupplierPartner data. The SupplierPartner Domain also includes data and contract operations related to the SupplierPartner Bills, disputes and inquiries.			
SupplierPartner ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	Example Secondary Business Process Framework Core Processes
Confirmation, requisition for Service Instance Problem reporting or <i>Initiated by the Supplier Partner</i> such as the proposal for a Wholesale Service Purchase, the quotation proposal, the quotation update proposal, the requisition confirmation the order executed and the problem report answer.			S/P Requisition Management
			S/P Problem Reporting & Management
			S/P Settlements & Billing Management
SupplierPartner Product The SupplierPartner Product ABE represents the Service Provider's knowledge of the Products that the Service Provider may order from the SupplierPartner	SCD&M – ILM	Supply Chain Development & Change Management	S/P Requisition Management
			Product & Offer Development & Retirement
			Service Development & Retirement
			Resource Development & Retirement
			Supply Chain Capability Delivery
SupplierPartner Problem Deals with problem associated with the Supplier Partner such as a request for technical assistance, Bill disputes etc.	S/PRM – A	S/P Problem Reporting & Management	Problem Handling
			Service Problem Management
			S/PRM Support & Readiness
			S/P Performance Management
SupplierPartner SLA Manages SLA Specification that contain standard components used to negotiate and define SLAs with SupplierPartners both for the Wholesale Service SP-SP and the End-Customer instances within the Wholesale service. The SupplierPartner SLA ABE also supports the negotiation and establishment of SLA instances. It also measures performance against the SLAs, and handles inquiries (both internal & external) about SLAs and the performance of	SCD&M – ILM and/or S/PRM – S&R	Supply Chain Capability Delivery*	S/P Problem Reporting & Management
		S/PRM Support & Readiness*	Supply Chain Capability Delivery

The SupplierPartner Domain includes all SupplierPartner-oriented data and contract operations associated with a SupplierPartner. Its scope encompasses, planning of strategies vs. SupplierPartners, handling of all types of contact with the SupplierPartner, the management of the relationship, and the administration of SupplierPartner data. The SupplierPartner Domain also includes data and contract operations related to the SupplierPartner Bills, disputes and inquiries.			
SupplierPartner ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	Example Secondary Business Process Framework Core Processes
Supplier Partner service with respect to the Supplier Partner SLA. It also handles SLA violations and the notification of the appropriate parties regarding violations. The ABE also identifies discount reasons for SLA violations and notifies S/P Payment of the discount reasons. The ABE also reads the notifications from the SupplierPartner, analyse the data and identifies the right party to deal with the data and sends out a notification.			S/P Requisition Management
			S/P Performance Management
SupplierPartner Performance Reads the SupplierPartners performance data for each End-Customer instance running in the infrastructure of the Service Provider and for which performance parameters are established by the Supplier/Partner SLA ABE. Only data that lies above the agreed thresholds and that may jeopardise the Service Providers service to the End Customer is sent by the SupplierPartner. The performance data from the SupplierPartner is analysed and redirected to the correct Service Mgmt process/actor for them to decide on appropriate actions. Based on SLA violation persistence of supplier/partners, their relation with the organization is terminated.	S/PRM – A	S/P Performance Management	Service Problem Management
			Service Quality Management
			S/PRM Operations Support & Readiness
			Supply Chain Development & Change Management
			S/P Problem Reporting & Management
SupplierPartner Order Handles single SupplierPartner orders and supervises various types thereof. This includes the quotation and order for the SupplierPartner Wholesale Service as well as the quotation and order for the EndCustomer instances within the Wholesale service.	CRM – F and/or S/PRM – F	Order Handling^	Supply Chain Development & Change Management
		S/P Requisition Management^	S/P Interface Management
			S/PRM Support & Readiness
SupplierPartner Bill Monitors the collection process and initiates different actions	S/PRM – B	S/P Settlements & Billing	S/P Performance Management

The **SupplierPartner** Domain includes all SupplierPartner-oriented data and contract operations associated with a SupplierPartner. Its scope encompasses, planning of strategies vs. SupplierPartners, handling of all types of contact with the SupplierPartner, the management of the relationship, and the administration of SupplierPartner data. The SupplierPartner Domain also includes data and contract operations related to the SupplierPartner Bills, disputes and inquiries.

SupplierPartner ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	Example Secondary Business Process Framework Core Processes
for overdue bills. It also handles credit violations. The ABE also provides the functionality to register disputes and adjustments on individual charges, invoices, and accounts.		Management	Service Guiding & Mediation
SupplierPartner Bill Inquiry The Supplier Partner Bill Inquiry ABE monitors the bill Disputes towards the Supplier Partner.	S/PRM – B	S/P Settlements & Billing Management	S/P Performance Management S/PRM Support & Readiness
Supplier Partner Payment This ABE monitors the payment process versus the Supplier/Partner.	S/PRM – B	S/P Settlements & Billing Management	S/P Performance Management
SupplierPartner Statistic The Supplier/Partner Statistics ABE represents the usage analysis of services purchased from the SupplierPartner, SupplierPartner profitability statistics and the overall performance of the Supplier/partner.	S/PRM – A	S/P Performance Management	S/P Settlements & Billing Management S/PRM Support & Readiness
Notes: * Due to process duality ^ Due to ABE ambiguity			

3.2.0 Engaged Party Domain (Preliminary)

The **Engaged Party** Domain includes all data and contract operations associated with individuals or organizations that are of interest to an enterprise. It represents of all types of contact with the party, the management of the relationship, and the administration of party data. The Engaged Party Domain also includes data and contract operations related to the party bills for products, collection of payment, overdue accounts, and the billing inquiries and adjustments made as a result of inquiries.

Engaged Party ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	Example Secondary Business Process Framework Core Processes
Party Strategy The strategies and the planning of the business relation with the other parties with input from other ABEs, such as MarketSales, Party Performance and Competitor Analysis.		Party Strategy & Planning	

The Engaged Party Domain includes all data and contract operations associated with individuals or organizations that are of interest to an enterprise. It represents of all types of contact with the party, the management of the relationship, and the administration of party data. The Engaged Party Domain also includes data and contract operations related to the party bills for products, collection of payment, overdue accounts, and the billing inquiries and adjustments made as a result of inquiries.			
Engaged Party ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	Example Secondary Business Process Framework Core Processes
Party The Party ABE represents the abstract concept of organization or individual that can play varying roles during interactions with an enterprise. Roles include customer, supplier/partner, employee, and so forth.		Party Engagement Development & Retirement	
Additional Party Entities Represents entities that are similar to those in the Customer ABE, such as PartyAccount and PartyAccountContact. This ABE was added to maintain backward compatibility and conformance of the Party ABE rather than include these entities within it.		Party Engagement Development & Retirement	
Party Interaction Represents communications with parties, and the translation of requests and inquiries into appropriate “events” such as the creation of an order, the creation of a bill inquiry, or the creation of a problem.		Party Interaction Management	
Party Statistic Represents the analysis of party usage patterns, profitability statistics.		Party Support & Readiness	
Party Problem Focuses on technical assistance and problem handling reported to and by other parties.		Party Problem Reporting & Management	
Agreement Represents a contract or arrangement, either written or verbal and sometimes enforceable by law, such as a service level agreement or a customer price agreement. An agreement involves a number of other business entities, such as products, services, and resources and/or their specifications.		Party Sourcing Management	
Party Service Level Agreement A Service Level Agreement ABE with one or more other parties. See the Agreement ABE in the in the Engaged Party domain for additional details.			

The Engaged Party Domain includes all data and contract operations associated with individuals or organizations that are of interest to an enterprise. It represents of all types of contact with the party, the management of the relationship, and the administration of party data. The Engaged Party Domain also includes data and contract operations related to the party bills for products, collection of payment, overdue accounts, and the billing inquiries and adjustments made as a result of inquiries.			
Engaged Party ABEs	Primary Vertical Business Process Framework Process Groupings	Primary Business Process Framework Core Processes	Example Secondary Business Process Framework Core Processes
Party Order Handles party orders and the various types thereof, such as regulated and non-regulated orders.		Party Order Handling	
Party Bill The Party Bill ABE represents the format of a bill, schedule the production of bills, party invoicing profiles, and credits and adjustments to bills.		Party Settlements & Payments Management	
Applied Party Billing Rates Deals with the correlation of related usage and rating, rates applied to the usage (both regulated and non-regulated), discounts to usage, and any taxes due on the rated usage.		Party Settlements & Payments Management	
Party Bill Collection Handles credit violations, actions for overdue debts, and facility billing audits.		Party Settlements & Payments Management	
Party Bill Inquiry Represents invoice inquiries associated with invoices received from other parties and handles disputes and adjustments on individual charges, invoices, and accounts.		Party Bill Inquiry Handling	
Agreement The Agreement ABE represents a contract or arrangement, either written or verbal and sometimes enforceable by law, such as a service level agreement or a customer price agreement. An agreement involves a number of other business entities, such as products, services, and resources and/or their specifications.			

3.2.1 Common Business Entities Domain

Common Business Entities were not mapped to Business Process Framework processes since these entities use is spread across a number of Business Process Framework processes.

Domain Name	Common Business Entities
Description	The Common Business Entities Domain represents business entities shared across two or more other domains. As such, these business entities are not “owned” by any particular domain. In some cases a common business entity represents a generic abstraction of other real-world business entities. For example, Business Interaction is an abstraction (super-class) of business entities such as Customer Order and Supplier/Partner SLA.
Aggregate Business Entity	
ABE	Location The Location ABE represents the site or position of something, such as a customer’s address, the site of equipment where there is a fault and where is the nearest person who could repair the equipment, and so forth. Locations can take the form of coordinates and/or addresses and/or physical representations.
ABE	Business Interaction The Business Interaction ABE represents an arrangement, contract, or communication between an enterprise and one or more other entities such as individuals and organizations (or parts of organizations). Interactions take on the form of requests, responses, and notifications.
ABE	Policy Policy consists of a set of layered ABEs that define specifications (for example, templates) and definitions of Policy entities that can be used in managing the behavior and definition of entities in other Domains. Policy takes three primary forms. The first is the definition of how policy is used to manage the definition, change, and configuration of other entities. The second is the definition of how policy itself is managed. The third is how applications use policies to manage entities.
ABE	Project The Project ABE represents the tools used by project managers to ensure that enterprise objectives of quality, cost, and time are achieved by planning and scheduling work. It uses common industry definitions of Project, Work Breakdown Structure and Activity to provide support to project managers.
ABE	Calendar The Calendar ABE represents Entities used to provide time related functions. This includes scheduling, time conflicts and time based presentation support.
ABE	Metric The Metric ABE defines a pattern to standardize the description of Metrics. It’s in vision that other projects use and specialize the Metric ABE in some way (creating sub-classes and adding new entities) such as PM (Performance Management), RA (Revenue Assurance), SLA (Service Level Agreement), etc.
ABE	Configuration A Configuration (also referred to as a Profile) defines how a Resource, Service, or Product operates or functions. A Configuration may contain one or more parts (which is realized by using the Atomic/Composite pattern, but it is represented as a single entity – ConfigurationRelationship), and each part may contain zero or more fields. Each field may have attributes that are statically or dynamically defined. Some of these fields have fixed values, while others provide values from which a choice or choices can be made (e.g., using the EntitySpec/Entity and/or CharacteristicSpec/CharacteristicValue patterns).
ABE	Root

Domain Name	Common Business Entities
Description	The Common Business Entities Domain represents business entities shared across two or more other domains. As such, these business entities are not “owned” by any particular domain. In some cases a common business entity represents a generic abstraction of other real-world business entities. For example, Business Interaction is an abstraction (super-class) of business entities such as Customer Order and Supplier/Partner SLA.
	A set of common business entities that collectively serve as the foundation of the business view. This set of entities enables the entities in different domains of the SID Framework to be associated with each other, providing greater overall coherence to the information framework.
ABE	Base Types A small simple object, like money or a date range, whose equality isn’t based on identity.
ABE	User and Roles User and Roles represents types of users and roles and their involvement in the usage of products, services, and resources.
ABE	Usage An occurrence of employing a Product, Service, or Resource for its intended purpose, which is of interest to the business and can have charges applied to it. It is comprised of characteristics, which represent attributes of usage.
ABE	Capacity Capacity represents the ability to provide a measured capability of the network.
ABE	Trouble or Problem A description of a problem that can be shared between the service provider and the customer. Trouble or Problem is an indication that an entity (such as Resource, Service or Product) is no longer functioning according to the expected SLA.
ABE	Trouble Ticket Represents a record used for reporting and managing the resolution of Trouble or Problem.
ABE	Performance Represents a measure of the manner in which a Product and/or Service and/or Resource is functioning.
ABE	Catalog A catalog can be defined as a collation of items and arranging them in a particular manner based on the need. For example, a product catalog
ABE	Metric Defines standards of measurement including units and values and how the values are determined. It also includes thresholds used to evaluate the metric and the consequences of violating the thresholds.

4. Information Framework Snapshot

4.1. Introduction

This chapter provides a summary of the key Information Framework business entities representative of Aggregate Business Entities (ABEs) contained within each domain. Also shown in the figures are the entities' associations with key entities associated with ABEs in other domains. This demonstrates that, while ABEs are groups of closely related entities, the entities contained within a domain and an ABE within a domain are naturally related to entities in other domains and their respective ABEs. Each diagram shown in this chapter was taken from Information Framework addenda.

The figures include

- Party
- Location (Place)
- Interaction
- Agreement
- Calendar
- Project
- Policy
- Customer
- Customer Order
- Product, Service, Resource
- Product Offering
- Product
- Service
- Resource

The associations depicted in the diagrams in this chapter are those currently identified by Information Framework development to date. As the Information Framework evolves further, additional associations may be found and current associations may be modified.

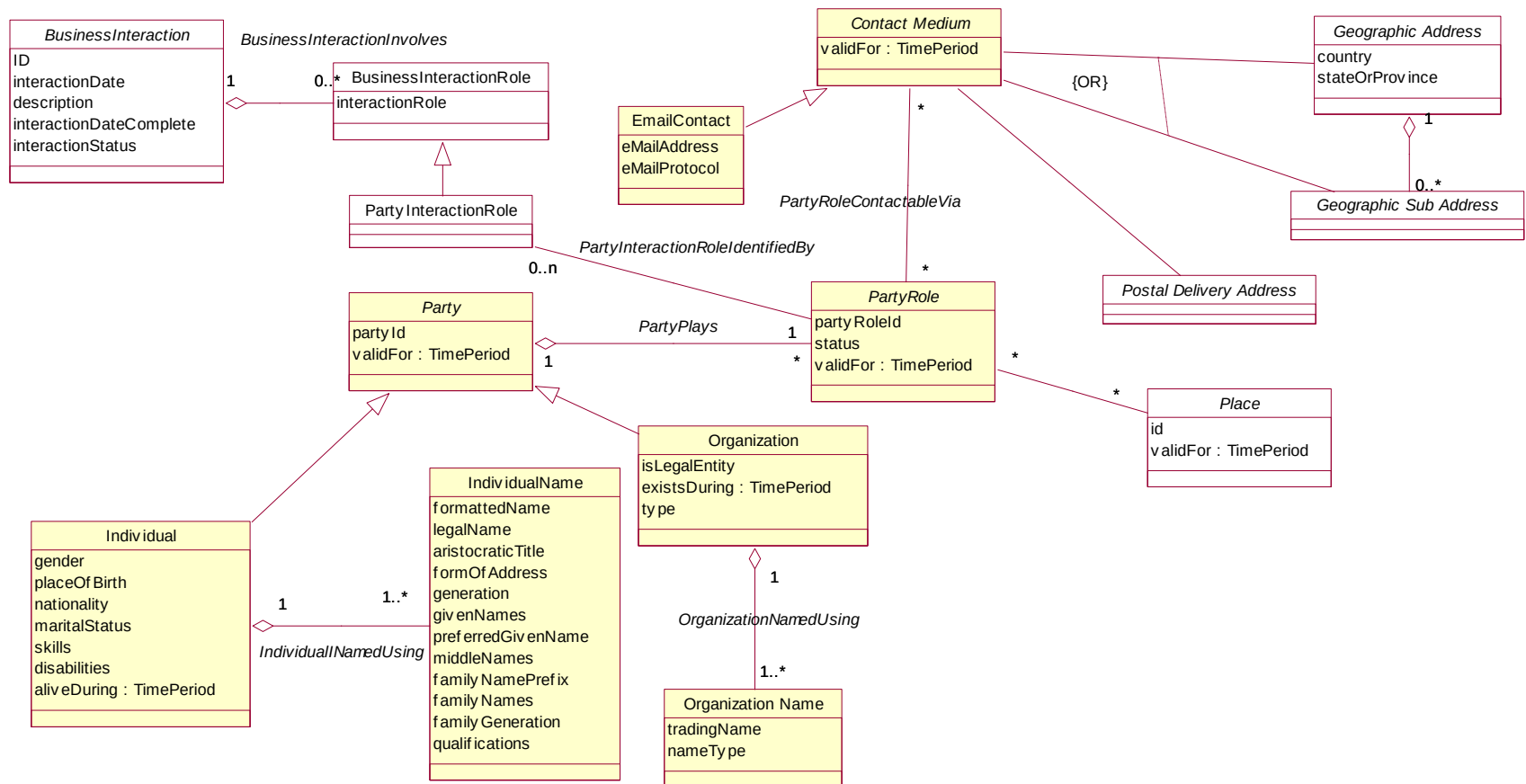
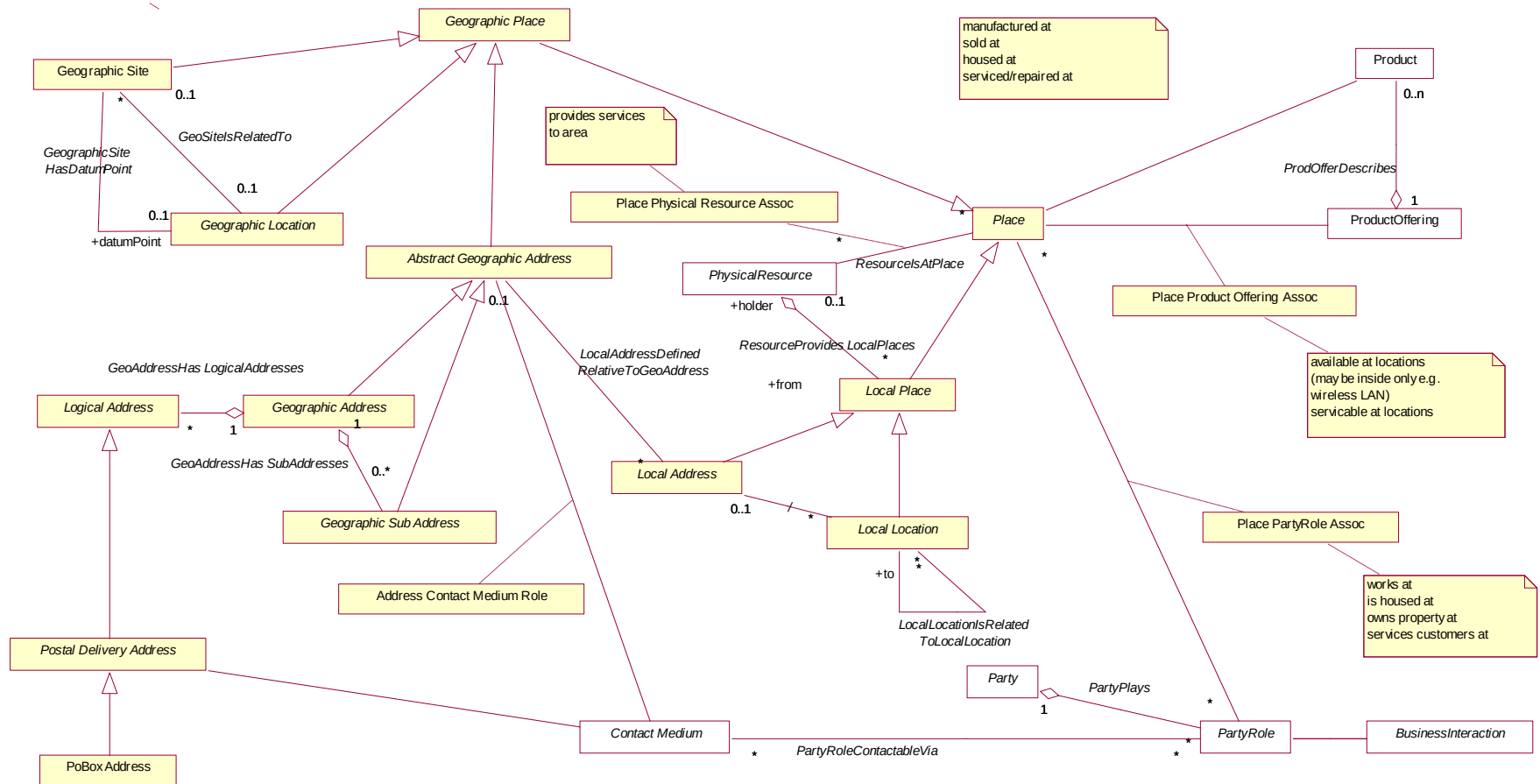


Figure 9 – Party and Related Entities



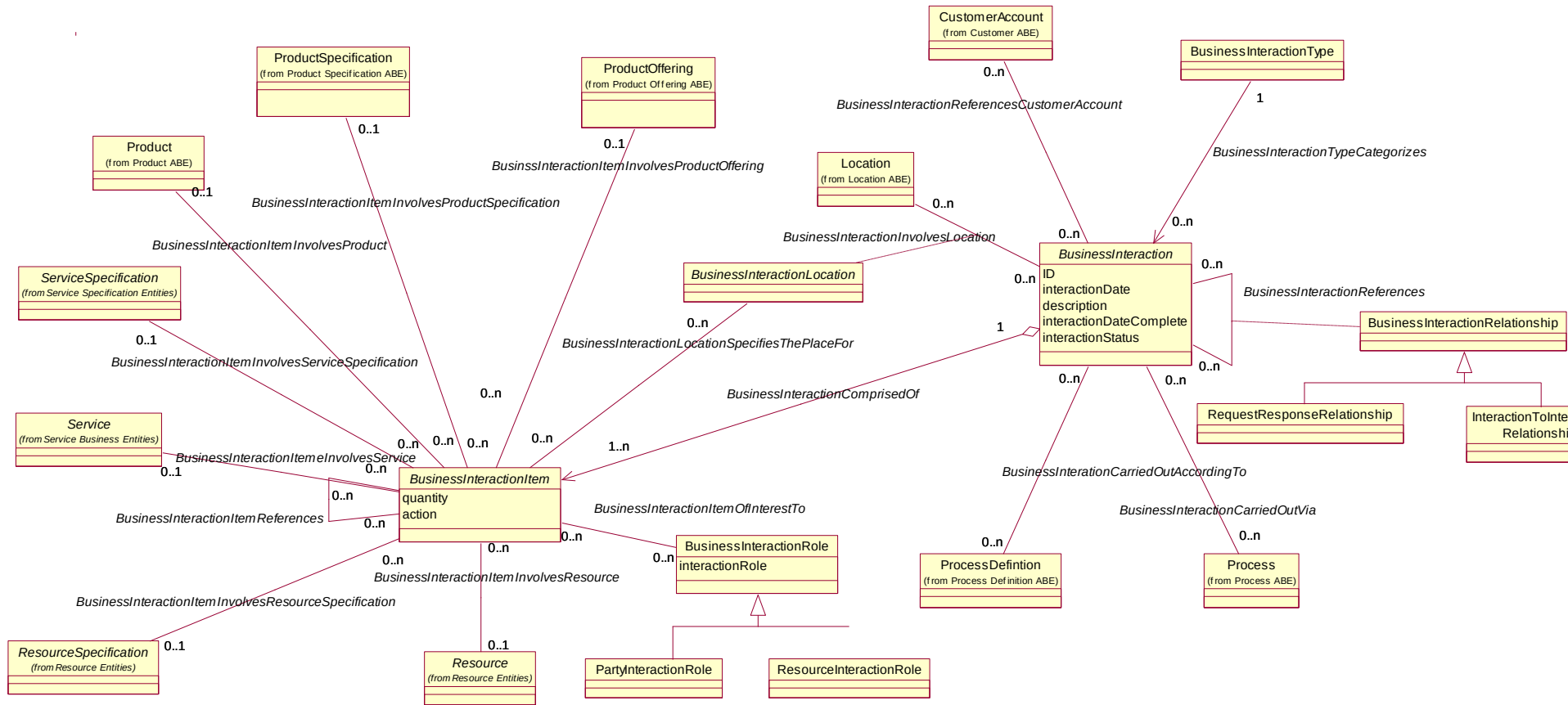


Figure 11 – Business Interaction and Related Entities

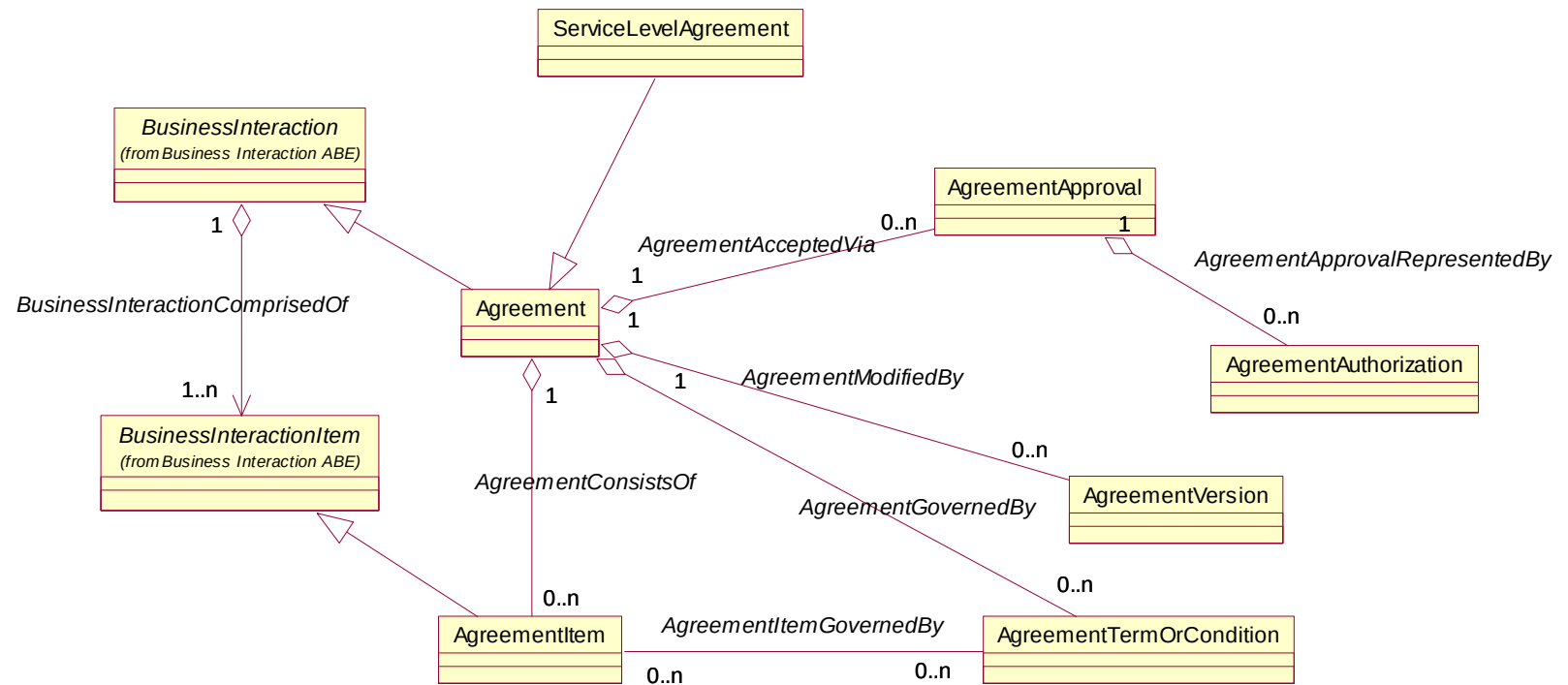


Figure 12 – Agreement and Related Entities

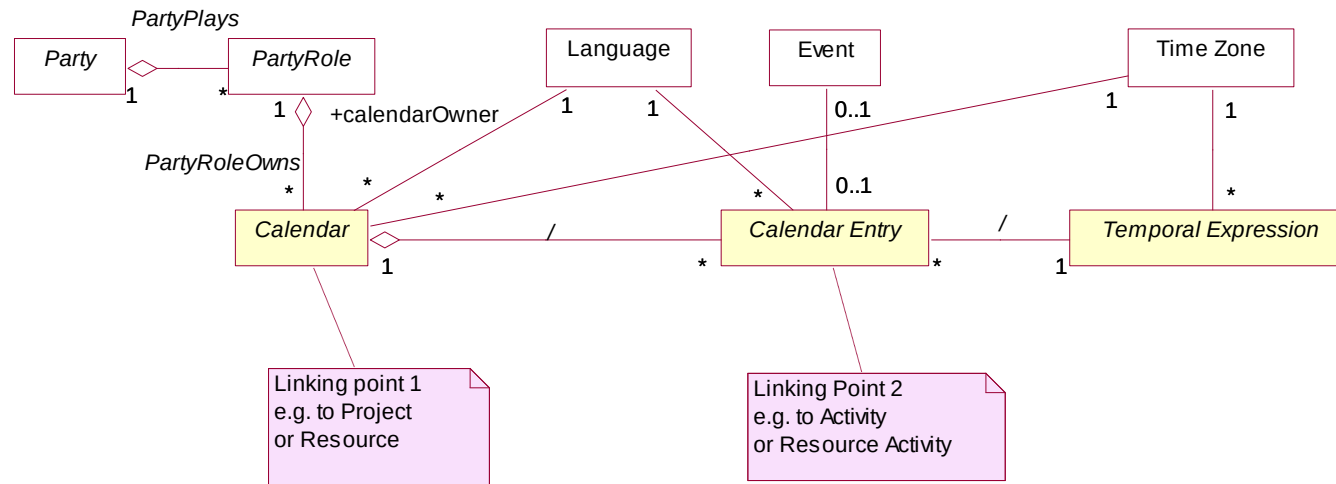


Figure 13 – Calendar and Related Entities

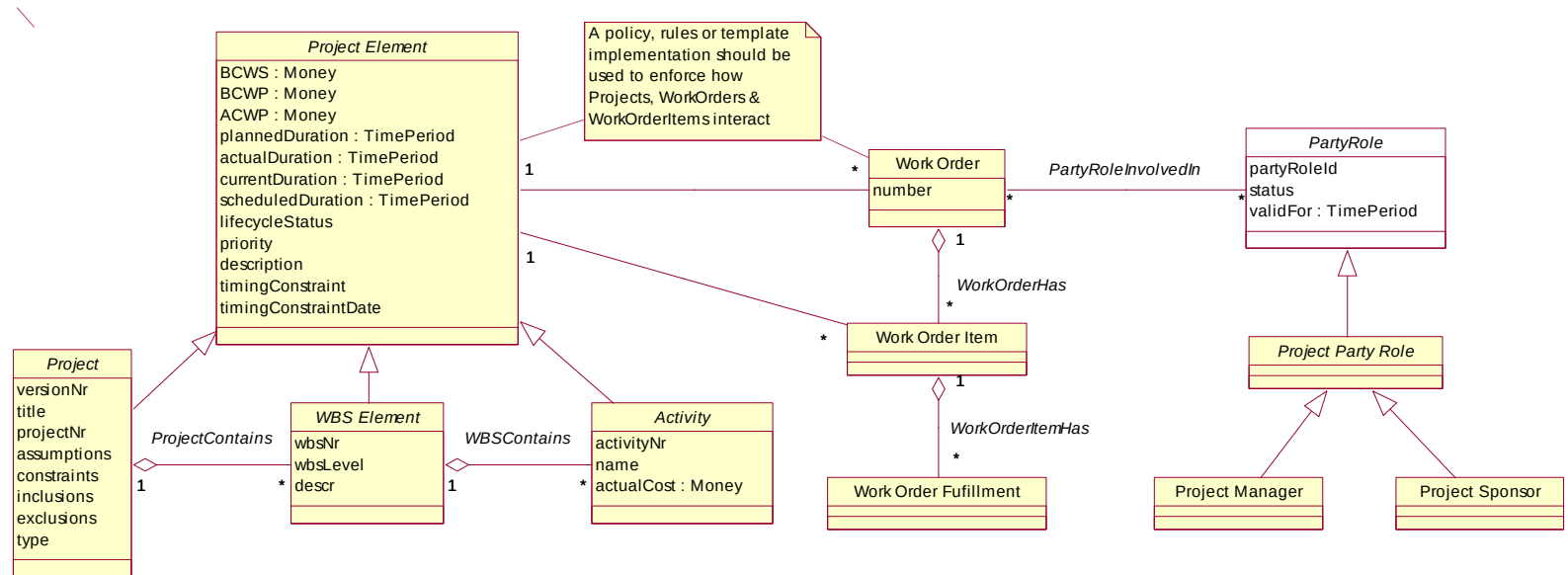


Figure 14 – Project and Related Entities

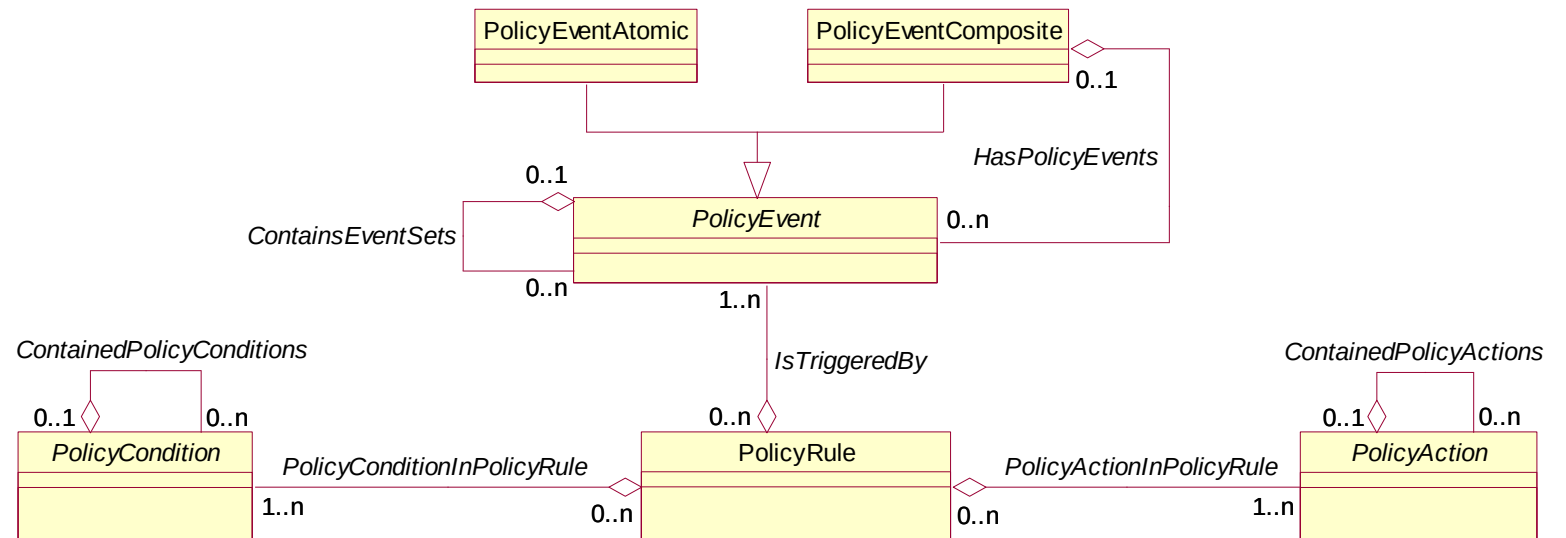


Figure 15 – Policy and Related Entities

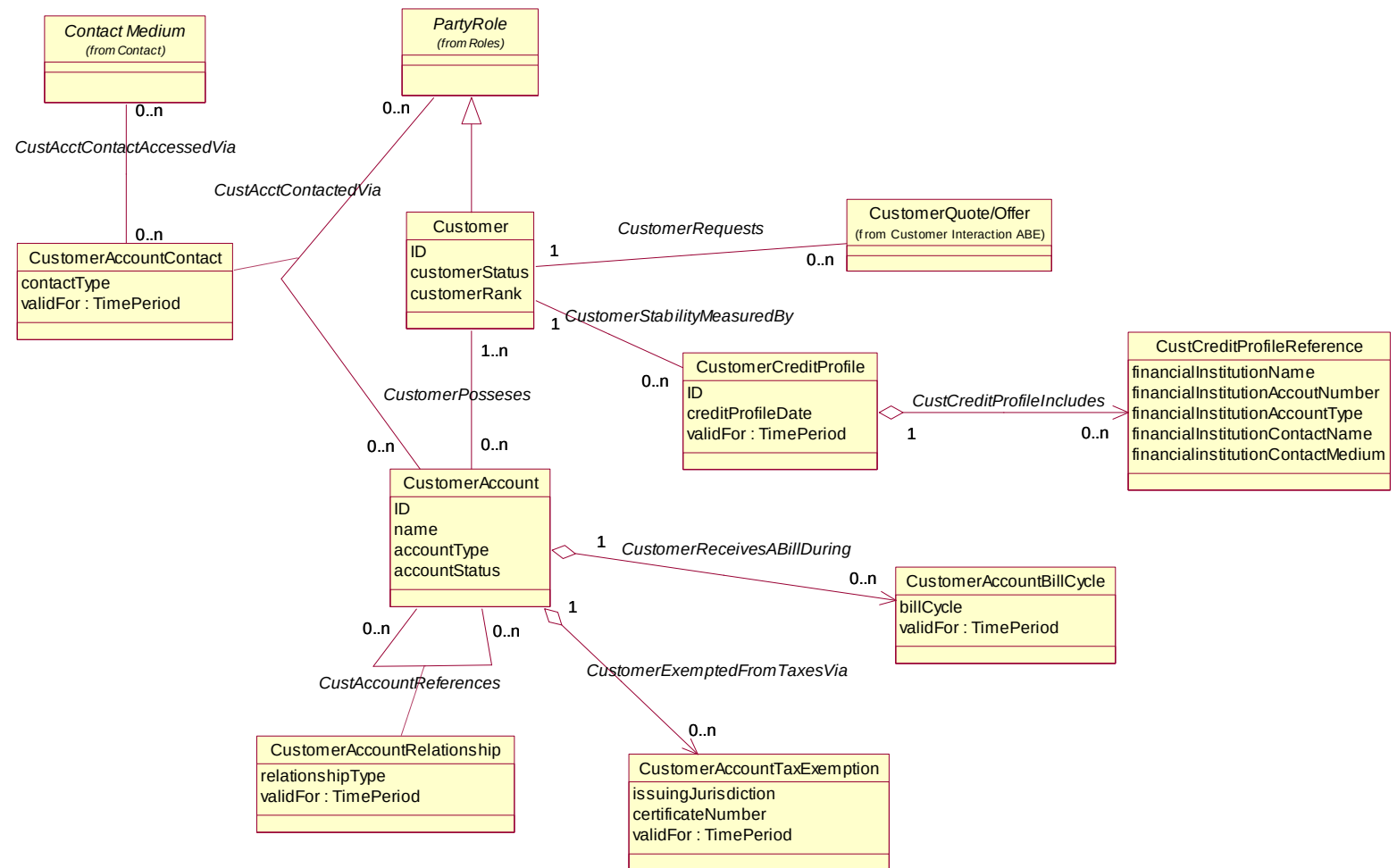


Figure 16 – Customer and Related Entities

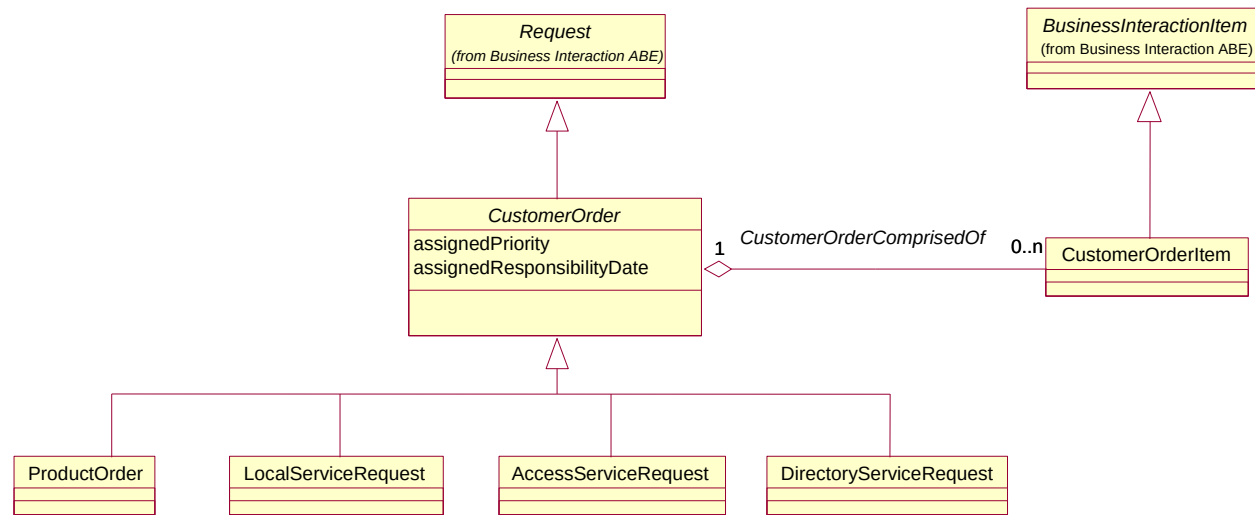


Figure 17 – Customer Order and Related Entities

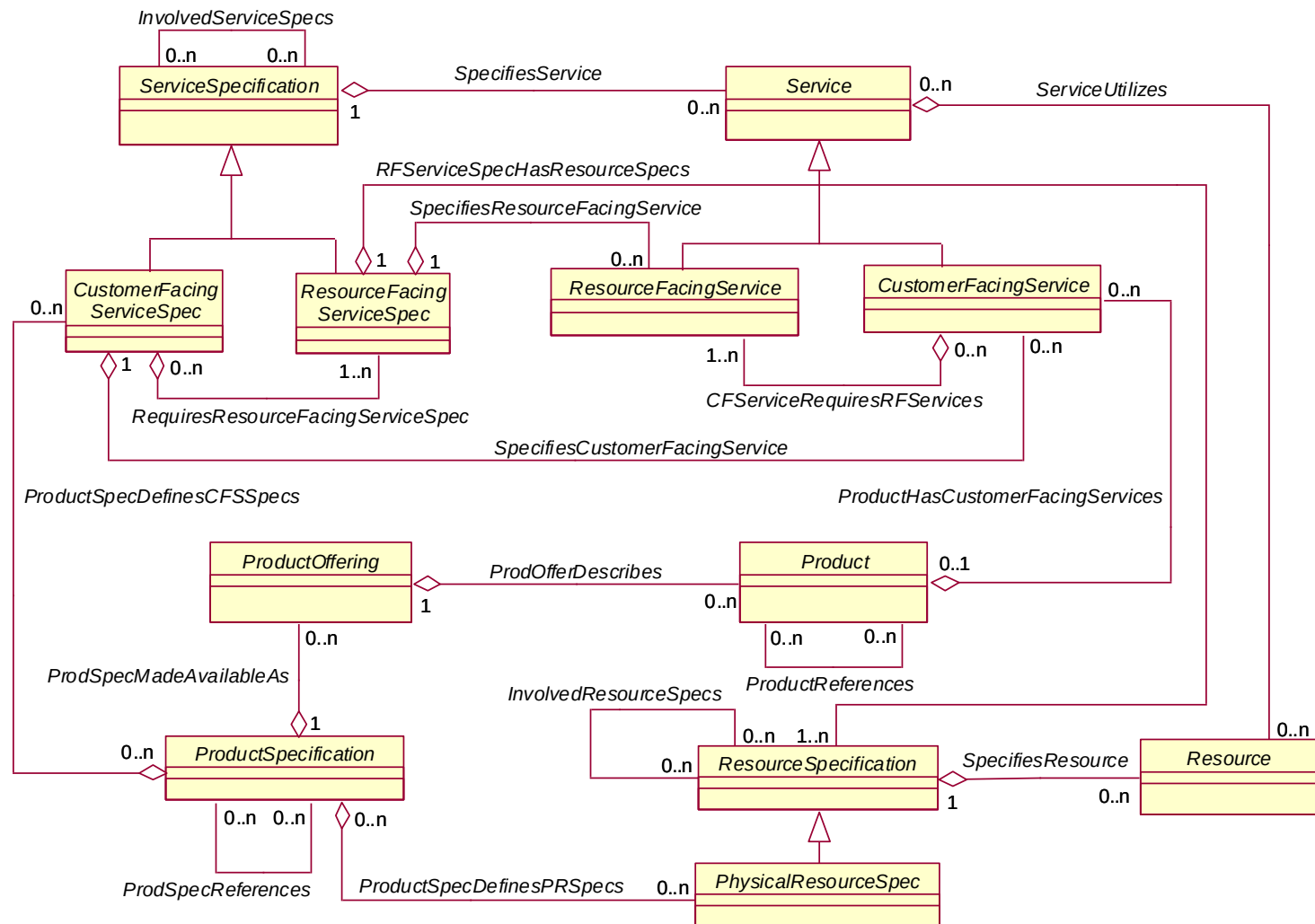


Figure 18 – Product, Service Resource Associations

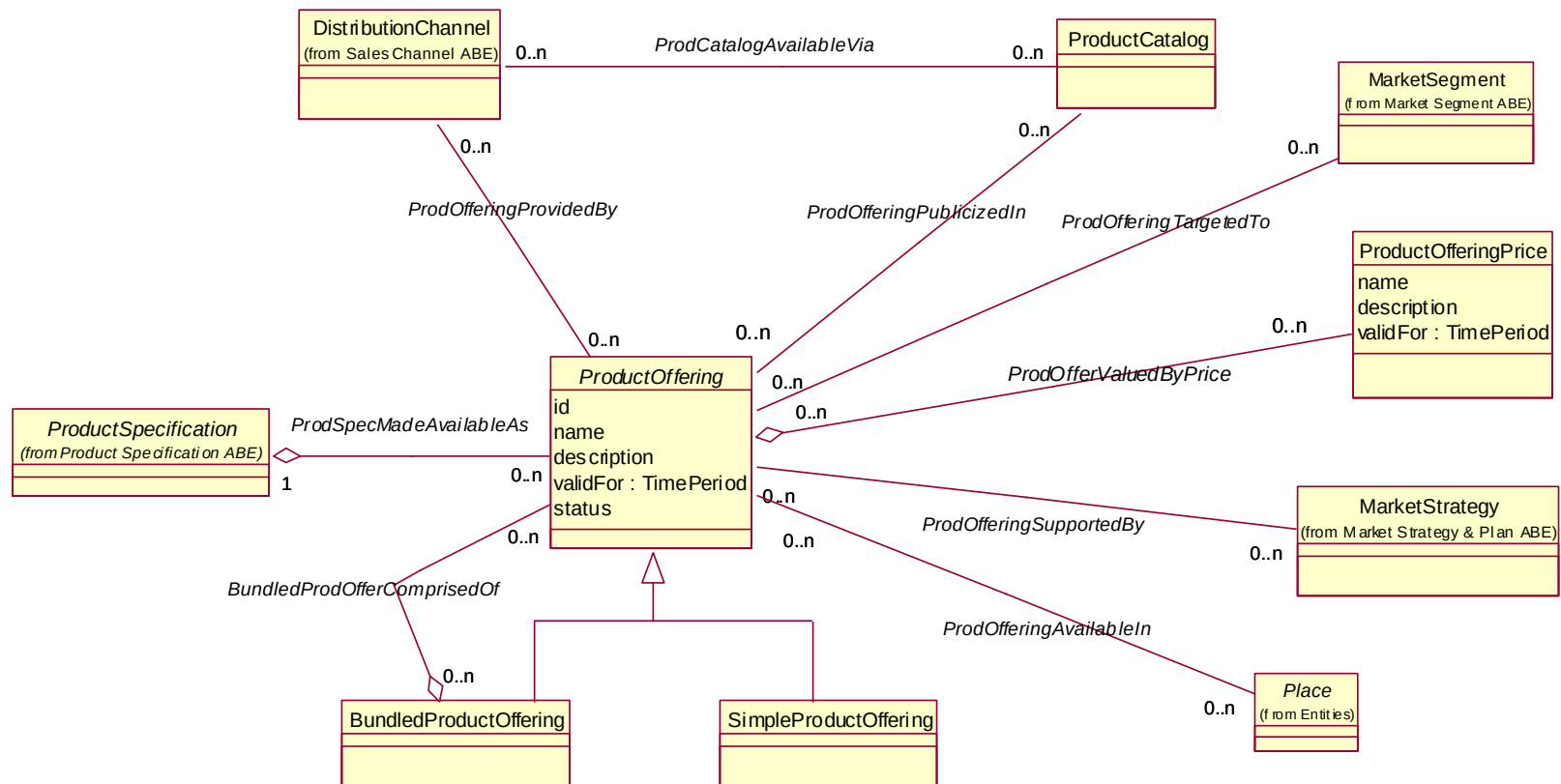


Figure 19 – Product Offering and Related Entities

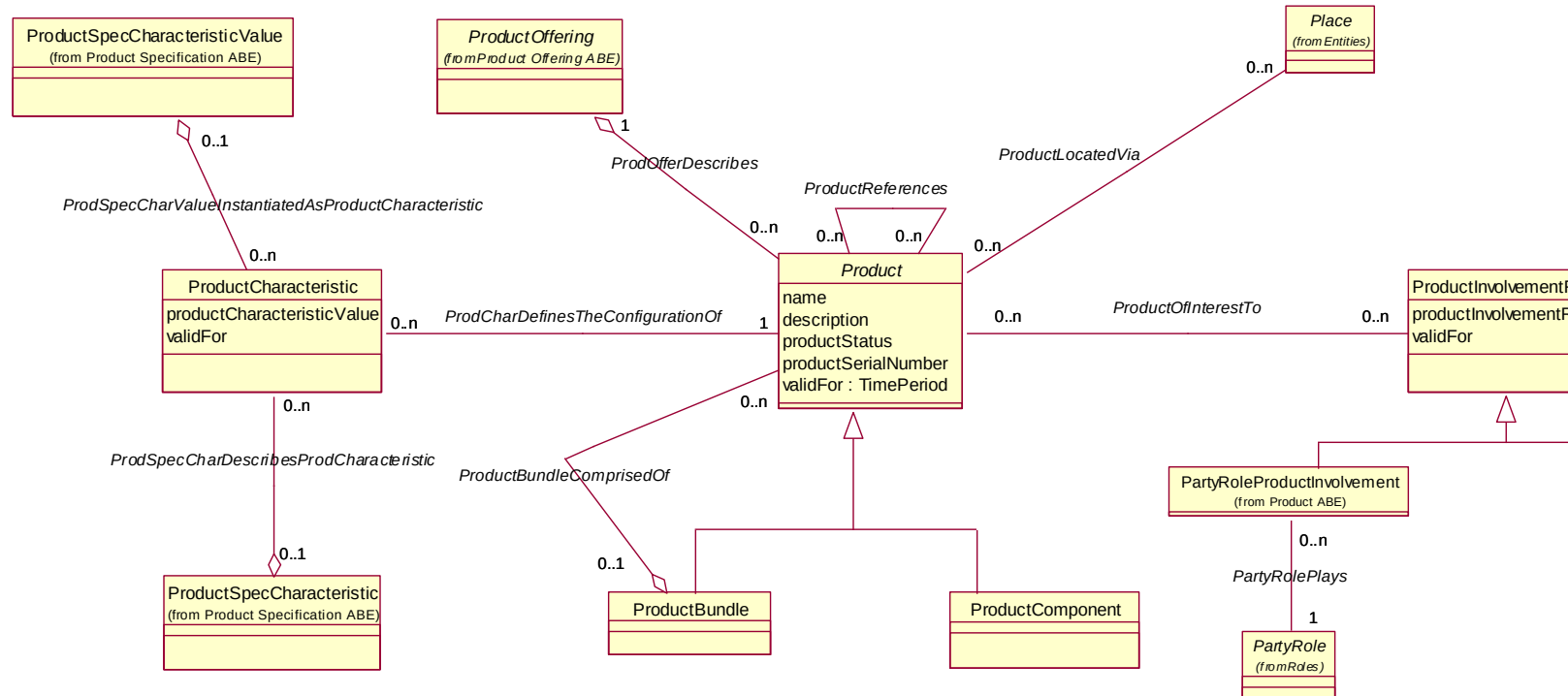


Figure 20 – Product and Related Entities

5. Annexes

5.1. Annex 1: Abbreviations and References

5.1.1. Abbreviations

OOAD	Object-Oriented Analysis and Design
OSS	Operational Systems and Software
UML	Unified Modeling Language

5.1.2. Non Normative References

The Business Process Framework(eTOM) – GB921 V7.0

6. Administrative Appendix

This Appendix provides additional background material about the TM Forum and this document. In general, sections may be included or omitted as desired; however, a Document History must always be included.

6.1. About this document

This is a TM Forum Guidebook. The guidebook format is used when:

- The document lays out a 'core' part of TM Forum's approach to automating business processes. Such guidebooks would include the Telecom Operations Map and the Technology Integration Map, but not the detailed specifications that are developed in support of the approach.
- Information about TM Forum policy, or goals or programs is provided, such as the Strategic Plan or Operating Plan.
- Information about the marketplace is provided, as in the report on the size of the OSS market.

6.2. Document History

6.2.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
0.1	Feb 2002		Initial sample of document format and content
0.5	May 2002		Updates based on review and feedback; this is the initial member review version
1.1	Oct 2002		Updates based on review and feedback from members.
1.2	Oct 2002		Further updates based on review by team members.
3.0	June 2003		Updates to Information Framework, added Information Framework/Business Process Framework mapping, inter-ABE diagrams

5.0	August 2004	John Reilly	Updated Business Process Framework/Information Framework mappings
6.0	July 2005	John Reilly	Updated with comments from Greg Fidler and Business Process Framework/Information Framework mappings
6.1	November 2005	Tina O'Sullivan	Converted to new template and corrected various administrative items.
6.2		Tina O'Sullivan	Updated Notice Statement & document status
7.5	January 2008	John Reilly	Updates for ITU-T submission
7.6	May 2008	Tina O'Sullivan	Made corrections prior to posting.
7.7	May 2009	Alicja Kawecki	Minor updates to reflect TM Forum Approved status
7.8	January 2010	Ken Dilbeck	Nameing changes per Marketing
7.9	March 2010	Pascale Pecha	Minor edits and correction of release number
7.10	March 2010	Alicja Kawecki	Minor cosmetic corrections for web posting
7.11	April 2010	John Reilly	Removed Service Applications ABE and changed Service & Instance Specific Rating to Service Guiding & Mediation
7.12	Apr 2010	Alicja Kawecki	Minor cosmetic corrections for web posting and ME
7.13	Jun 2010	Alicja Kawecki	Updated Notice
7.14	December 2010	John Wilmes	Removed reference to GB926, fixed CR 2092
7.15	March 2011	Alicja Kawecki	Minor formatting corrections prior to web posting and ME

7.16	September 2011	Alicja Kawecki	Updated to reflect TM Forum Approved status
7.2	January 2012	John Reilly	Updated SID Framework figures
7.3	March 2012	Alicja Kawecki	Minor formatting corrections prior to web posting and ME
7.4	October 2012	Alicja Kawecki	Updated to reflect TM Forum Approved status
7.4.1	November 2013	Alicja Kawecki	Applied rebranding, updated cover, header, footer & Notice
7.5.1	April 2014	John Reilly	Updated Framework figures and Domain tables.
7.5.2	April 2014	John Reilly	Corrected Framework Level 1 figure.
7.5.3	May 2014	John Reilly	Corrected ABE descriptions, moved Agreement ABE to Engaged Party domain.
7.5.4	May 2014	Alicja Kawecki	Updated footer, Notice; minor formatting edits prior to posting
14.5.0	October 2014	John Reilly	Update eTOM mappings and Framework diagrams
14.5.1	March 2015	Alicja Kawecki	Updated cover, header, footer and Notice to reflect TM Forum Approved status

6.2.2. Release History

Release Number	Date Modified	Modified by:	Description of changes
Release 7.5	March 2008	J. Reilly	For ITU-T submission
Release 8.1	March 2010	Ken Dilbeck	To reflect change from Solution Frameworks to Frameworks
Release 9.0	April 2010	J. Reilly	ABE update and eTOM mapping updates.
Release 9.5	December 2010	J. Wilmes	Updated Notice
Release 12	January 2012	John Reilly	Updated SID Framework figures
Release 14	April 2014	John Reilly	Updated Framework figures and Domain tables.
Release 14.5.0	October 2014	John Reilly	Update eTOM mappings and Framework diagrams

6.3. Acknowledgments

This document was prepared by the members of the TM Forum Information Framework (SID) team.